

Inverse Airfoil Design as Constrained Root-Finding: A Monolithic CST–Newton Formulation

Sharath Sathish

Independent Researcher

TechNektar, Indian Institute of Science

sharath.sathish@gmail.com

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Abstract

Inverse airfoil design, recovering a geometry that produces a prescribed surface pressure or edge-velocity distribution, is recast here as a single determined nonlinear root-find rather than an objective-function search. Parameterising the surface with Class-Shape-Transformation (CST) coefficients that enter the geometry linearly makes the geometric sensitivity of the surface exact, constant and design-independent. It also makes geometric design constraints, among them leading-edge radius, trailing-edge thickness and inscribed area, linear algebraic rows rather than nonlinear predicates. Appending the CST coefficients as unknowns to a coupled viscous/inviscid Newton solver (`mfoil`) therefore converts constrained shape optimisation into constrained root-finding: one square system, no outer loop, no surrogate. The architecture is validated with a falsifiable self-consistency test in which a known CST coefficient vector is recovered from its own self-generated target to $\|A - A^*\|_\infty = 2.75 \times 10^{-11}$ in six Newton iterations. The recovered geometry reproduces the reference section’s fully released (natural-transition) aerodynamic coefficients to $\Delta c_l = 3.4 \times 10^{-12}$. An ablation matrix identifies the primary uniqueness guard for the resulting square system: sensitivity-optimal (QR-pivoted) selection of target stations, not initial-guess quality, separates recovery of the true design from clean convergence to a spurious but residual-zeroing root. Measured against a competently-tuned nested Levenberg–Marquardt baseline under two independent fair-paired controls, the monolithic architecture requires $3.1\text{--}8.1 \times$ fewer counted flow solves and $3.4\text{--}3.5 \times$ less wall-clock time. This is a real but modest reduction, not the two-to-three-orders-of-magnitude headline hypothesised a priori, and it comes with determinism, an exact analytic Jacobian for the constraint rows, and per-iteration failure-mode diagnostics for which the nested baseline has no analogue. Generalisation is then evaluated on two pre-registered panels. A 20-section NACA panel recovers all 18 generable sections to $\|A - A^*\|_\infty \leq 1.51 \times 10^{-10}$; a 117-section panel drawn from the UIUC coordinate database recovers every one of its 83 converged sections to better than 10^{-4} , while missing the pre-registered composite criterion on iteration count rather than on accuracy. Both panel outcomes are reported alongside the exclusions they rest on and the geometric bias those exclusions carry. The architecture is finally placed on a comparison table against MISES’s own modal inverse mode, the nearest prior CST-based inverse method, and the current generative and learned-surrogate inverse-design literature, on formulation class, cost, determinism and constraint-handling grounds rather than a single flow-solve number, since the methods are not commensurable on that axis alone.

Keywords: inverse design; airfoil; class-shape transformation; Newton method; viscous–inviscid coupling; identifiability; constrained root-finding

1 Introduction

Inverse aerodynamic design, given a target surface pressure or isentropic Mach distribution, recovering the geometry that produces it, is classically posed either as a well-posed but restrictive conformal-mapping problem [Lighthill, 1945, Volpe and Melnik, 1986], or as an outer optimisation loop wrapped around a converged forward flow solve. The latter is the practical default in industry: a geometry parameterisation, a surrogate or gradient-based optimiser, and hundreds to thousands of fully converged flow solves per design.

The motivation for the present paper is autobiographical. The author’s own doctoral work on sCO₂ axial-turbine blading [Sathish et al., 2019] used exactly this pattern: Class-Shape-Transformation (CST) geometry, a B-spline control-point wrapper, and a surrogate-assisted optimisation loop against MISES. That work already used CST for blade parameterisation, together with geometric constraints encoding expert design practice, but it used them *inside* an optimiser, and its closing section named the open problem verbatim: integrating CST directly with a Newton-based flow solver to escape the nested loop. The present paper’s claim is that the linearity which made those constraints convenient there is strong enough to remove the optimiser altogether.

That escape route already exists in one solver. The modal inverse mode of MISES [Drela, 1990] appends mode-shape amplitudes directly to its global Newton system and drives them so that computed surface speed matches a target: a determined root-find, not a search. This is not new; Drela [1990] described it between 1986 and 1990. What the built-in mode shapes of MISES are *not*, however, is a complete, geometrically meaningful, analytic design basis. They are heuristic bump functions, not a family whose coefficients double as leading-edge radius, trailing-edge thickness or inscribed area. Kulfan’s CST parameterisation [Kulfan, 2008] is such a basis, and it is one of the analytic families examined in the reference geometric benchmark of Masters et al. [2017]. CST has also already been coupled to inverse design once, by Lane and Marshall [2010]. There, CST is used to *smooth* a pressure-residual-driven normal-direction shape correction inside an outer iterative loop; CST is the smoother, not an unknown in a Newton system. The present work does not repeat either of these. It combines a property of CST that neither prior use exploits.

1.1 Related work

Classical inverse design. Inverse airfoil design predates any of the parameterisations discussed below. Lighthill’s conformal-mapping method [Lighthill, 1945] is the founding well-posed formulation: a target speed distribution on the unit circle is mapped conformally to the physical airfoil. Three integral closure conditions, closure of the contour, a fixed leading-edge radius, and a prescribed circulation, are required for a solvable, self-consistent shape to exist at all. This closure-condition structure is the direct conceptual ancestor of Section 2.2’s linear rows: both are answers to the same question, what makes an arbitrarily chosen target realisable by *some* member of the design family, asked eighty years apart under different geometric representations. Volpe and Melnik [1986] sharpened well-posedness for the transonic regime, showing that an inverse problem posed with pressure rather than speed as the target, and with the wrong closure conditions, is generically ill-posed. Existence and uniqueness are not automatic once the target is chosen independently of the geometry family, a caveat that motivates the identifiability discussion of Section 5.1 in a different, discrete and finite-dimensional setting. Eppler and Somers’s design method [Eppler and Somers, 1980] and Selig and Maughmer’s PROFOIL [Selig and Maughmer, 1992] are the mature, still-used engineering descendants. Both are multi-point conformal-mapping inverse solvers that target several operating conditions’ pressure distributions simultaneously by blending mode functions, and both remain outside the Newton-coupled, viscous-inclusive family this paper and MISES belong to. They

close the inverse problem analytically at the potential-flow level, then couple to a boundary-layer method non-simultaneously, rather than solving flow and geometry as one system.

Parameterisation comparison. Sobieczky’s PARSEC [Sobieczky, 1999] parameterises an airfoil by twelve directly-meaningful quantities, among them leading-edge radius, the thickness and camber extrema and their locations, and the trailing-edge angle and thickness, fitted by a polynomial basis. It is geometrically transparent in the same spirit as CST’s endpoint identities (Eq. (6)), but through a basis whose coefficients are not independent design freedoms in the same linear sense: PARSEC’s basis functions depend on the target values themselves, since the fit is re-solved per target, whereas CST’s Bernstein basis $S_i(\psi)$ is fixed once n is chosen, independent of what shape is being represented. This is the property Eq. (5) depends on. The benchmark of Masters et al. [2017] is the field’s reference comparison: over two thousand airfoils and seven parameterisation families, including CST, PARSEC, B-splines, Hicks–Henne bumps, a radial-basis domain-element method, and orthogonal modes derived by singular-value decomposition of a training database. Its headline finding is that twenty to twenty-five design variables suffice for all of the families tested, with the methods performing comparably at that budget and the data-driven modal method marginally the most efficient. That last qualification is the one this paper turns on. Data-driven modes fit an existing database efficiently at the cost of needing that database, and they carry no closed-form geometric-functional structure of the kind Section 2.2 exploits. Among the analytic families that do, CST is a well-supported choice, and it is the one whose particular algebraic property, linearity in the coefficient vector A , this paper’s contribution depends on.

Newton-coupled viscous/inviscid solvers. The architecture this paper appends CST to is the ISES lineage of Drela and Giles [1987]: a two-equation integral boundary-layer closure coupled *simultaneously*, not iteratively, to a panel-method inviscid solution through a single global Newton system, with the e^n transition model providing the closure that makes laminar-to-turbulent transition a smooth part of that system rather than a switched boundary condition. XFOIL [Drela, 1989] is the widely used low-speed member of that lineage, and `mfoil` [Fidkowski, 2022] is a modern, open, from-scratch reimplement of exactly this architecture: the ISES and XFOIL viscous–inviscid coupling without XFOIL’s Fortran legacy code, in Python, with an exposed sparse global Jacobian. This is why it serves as the first testbed here (Section 2): the extended system of Section 2.3 could not be built without an analysis code that already exposes its own $\partial R/\partial U$ block for direct augmentation. The parametric inverse mode of MISES was, moreover, designed from the outset to accept an arbitrary geometry-definition system through a black-box interface [Drela, 1990], so the mechanism a later transplant of this architecture would use already exists in the target solver, and has never been exercised with a CST-as-unknowns formulation.

Modern learned inverse-design methods. A distinct, much newer line of work learns the pressure-to-geometry map directly from data rather than solving a physics-based system at all. Conditional generative adversarial networks [Yilmaz and German, 2020] and, more recently, diffusion models learn a stochastic generative map from a target pressure or aerodynamic-coefficient vector to a geometry, trained offline on a large precomputed database, with inference reduced to a single or few-step forward network pass. A classifier-free-guided denoising diffusion model reports a 34.4 percent precision improvement over Wasserstein-GAN baselines on an airfoil-generation benchmark [Deng et al., 2025]. A hybrid approach trains a physics-informed surrogate on low-fidelity viscous–inviscid-coupled flowfields of the same class this paper’s solver produces, and reports at least an order-of-magnitude online inference speedup relative to direct analysis, again after an offline training

investment [Wong et al., 2024]. These methods answer a different question well: given a large library of prior designs and target statistics resembling that library, produce a plausible geometry near-instantly. They do not provide a convergence guarantee, an exact constraint-row Jacobian, or a certificate that the returned geometry actually reproduces the requested pressure distribution to any specified tolerance. The network’s output is evaluated *after* the fact by running a flow solver on it, not constrained *during* generation the way Eq. (10)’s equality rows are. Section 5.9’s comparison table places this paper’s architecture and the learned-inverse family on common columns explicitly, because the two are not competing solutions to the same technical problem so much as different points on an accuracy-guarantee against amortised-cost trade surface.

Turbomachinery inverse design. Porta et al. [2026] benchmark Kolmogorov–Arnold networks against multilayer-perceptron and Gaussian-process regressors for inverse turbine-passage design, trained on approximately thirty thousand blade profiles generated by an automated MISES-coupled optimisation pipeline over inlet flow angles from -50° to 0° and outlet angles from 50° to 75° . The trained network infers a loading-consistent geometry in about 0.1 seconds with a mean outlet-angle error of 0.086° . Structurally this is the same learned-surrogate pattern as the previous paragraph, specialised to cascades, and it is the closest published turbomachinery analogue of this paper’s motivating problem. It is, however, once more amortised-cost statistical inference over a precomputed design database, not a per-target root-find with a convergence certificate. Their pipeline includes a predicted-error feasibility proxy that flags statistically ill-posed inverse targets before training-set generation. This is the statistical-learning analogue of this paper’s realisability guard (Section 2.7), independently motivated by the same underlying problem, that some targets are simply not achievable by the design family in use, but answered by a learned classifier rather than a closed-form linearisation.

1.2 The contribution and its scope

CST’s surface parameterisation

$$\zeta(\psi) = C(\psi) \sum_i A_i S_i(\psi) + \psi \zeta_T \quad (1)$$

is *linear* in the coefficient vector A , and two consequences follow. Neither is available to a generic geometry representation, such as a B-spline with a control-point wrapper, once local-frame perturbations, leading-edge tangent-angle displacements or re-fitting are introduced to reach the same design freedom. First, the geometric sensitivity $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ is exact, constant and design-independent, so any operator built on it is assembled once rather than re-derived inside a Newton loop. Second, the geometric functionals that matter for manufacturability, among them leading-edge radius, trailing-edge thickness, wedge angle, inscribed area and leading-order curvature continuity, are themselves *linear* in A , so they enter the Newton system as constant-coefficient algebraic rows rather than non-smooth predicates or penalty terms. The contribution of this paper is not geometry unknowns in a Newton system, which is Drela’s, and not CST used somewhere inside inverse design, which is Lane and Marshall’s. It is the combination: an analytic basis whose linearity makes the geometric sensitivity exact and constant, *and* whose geometric functionals make design constraints linear rows, together converting constrained shape optimisation into constrained root-finding, one square Newton system, solved once, with no outer loop.

Two scope limits on that claim are stated here rather than left for the reader to discover. The first concerns which half of the claim the reported experiments exercise. Every configuration reported in this paper runs either with the leading edge prescribed, in which case no constraint row is assembled

at all, or with a single shared-leading-edge row; the trailing-edge-wedge and inscribed-area rows are implemented and unit-tested against independent quadrature but are not exercised by any measured result (Section 2.2). The second concerns the Jacobian. Eq. (5)’s exact constant sensitivity is used directly in the constraint rows and in the linear pre-solve of Section 2.7; the flow block $\partial R/\partial A$, by contrast, is obtained by finite differences over A , because the vendor solver exposes no complete geometric sensitivity of its own residual (Section 2.4). Both limits are consequences of what an unmodified third-party solver makes available, and both are revisited in Section 7.3.

1.3 The corrected headline

A priori, the architecture was hypothesised to require two to three orders of magnitude fewer flow solves, plus determinism and uniqueness, relative to a nested-optimisation baseline. That number was never measured against a competent baseline, and it is not repeated here. A controlled ablation study (Section 5) instead measures the architecture against a modern, properly-scaled, tightly-toleranced Levenberg–Marquardt solver [Virtanen et al., 2020], and finds 3.1 to $8.1\times$ fewer counted flow solves depending on initialisation strategy, under two independent fair-paired controls. Because the counting currency is itself contestable, Section 5.7 reports the same two pairings in measured wall-clock time, where the advantage narrows to $3.4\text{--}3.5\times$; the manuscript treats the wall-clock figure, not the solve count, as the conservative statement of the result. The measured ratio comes with qualitative properties no cost figure captures: determinism, with no stochastic restarts; an exact analytic Jacobian for whatever constraint rows are active, free of finite-difference noise; and a battery of per-iteration diagnostics (Section 3) that discriminate among the architecture’s failure modes and for which the nested baseline has no equivalent. The paper’s two central empirical results are therefore the correction of the a priori hundred-fold claim and the discovery of a station-selection identifiability mechanism that the original hypothesis did not anticipate (Section 5.1).

1.4 Structure of the paper

Section 2 states the CST mathematics, the constraint rows, the extended Newton system and its degree-of-freedom accounting, the derivative strategy the target solver’s exposed interface actually forces, and the two solver-specific closures, forced transition and the realisability pre-solve, needed to make the Newton iteration well-posed. Section 3 catalogues the four failure modes the architecture is vulnerable to and the diagnostics built to discriminate between them. Section 4 describes the self-consistency test that falsifies or confirms the hypothesis outright. Section 5 reports the ablation matrix, the identifiability finding, the Bernstein-conditioning cliff, the two panel evaluations, the corrected cost comparison and the observed convergence behaviour. Section 6 describes the released software artifacts, an interactive application and a project site, through which the method can be exercised without installing anything. Section 7.1 draws out the common structure of the four solver-specific mitigations, Section 7.2 states the study’s threats to validity and Section 7.3 its scope limits, and Section 7.4 discusses why the pre-registered cost threshold was missed. Section 8 closes with the cascade and MISES extensions this architecture was designed to scale into.

2 Formulation

2.1 CST parameterisation and its linearity

Let $\psi = x/c \in [0, 1]$ and $\zeta = z/c$. The CST class function for a round-nose, sharp-tail family is

$$C_{N_1}^{N_2}(\psi) = \psi^{N_1}(1 - \psi)^{N_2}, \quad N_1 = 0.5, \ N_2 = 1.0, \quad (2)$$

and the shape function is a Bernstein-polynomial expansion of order n per surface,

$$S_i(\psi) = K_i \psi^i (1 - \psi)^{n-i}, \quad K_i = \binom{n}{i}, \quad S_u(\psi) = \sum_{i=0}^n A_{u,i} S_i(\psi), \quad S_l(\psi) = \sum_{i=0}^n A_{l,i} S_i(\psi). \quad (3)$$

The upper and lower surfaces are the per-surface form of Eq. (1),

$$\zeta_u(\psi) = C(\psi) S_u(\psi) + \psi \zeta_{T,u}, \quad \zeta_l(\psi) = C(\psi) S_l(\psi) + \psi \zeta_{T,l}, \quad (4)$$

with $\zeta_{T,u} = +t_{TE}/2c$ and $\zeta_{T,l} = -t_{TE}/2c$, both already non-dimensionalised by chord along with ζ itself. The sign convention is binding throughout: lower-surface coefficients are stored with their natural negative sign, so the shared-leading-edge row reads $A_{u,0} + A_{l,0} = 0$.

Figure 1 shows the construction term by term. The class function of panel (a) is fixed geometry that carries no dependence on A at all; the Bernstein shape functions of panel (b) are fixed polynomials; and the surface of panel (d) is the A -weighted sum of the terms in panel (c). Only the choice of N_1 and N_2 is nonlinear, and those are held fixed throughout this work at the round-nosed, sharp-trailing-edge values $N_1 = \frac{1}{2}$, $N_2 = 1$.

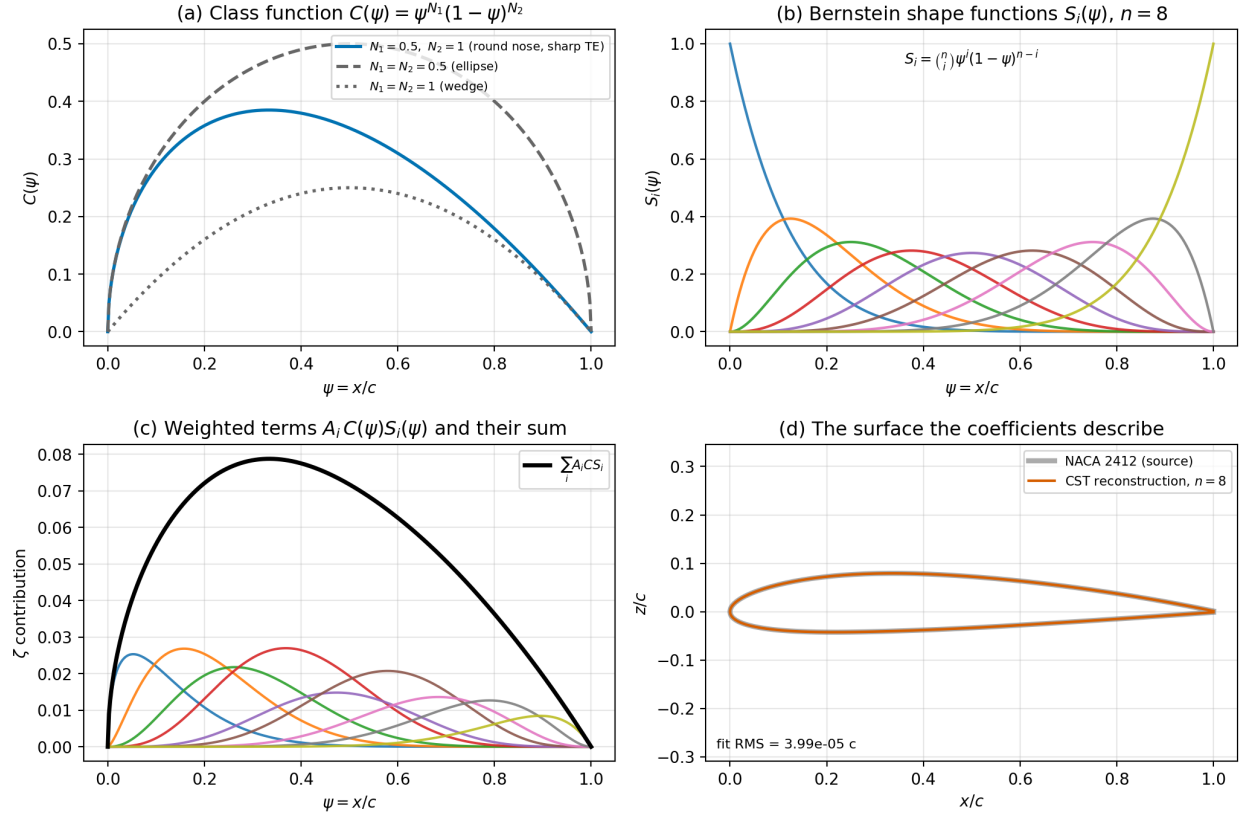


Figure 1: The CST construction, evaluated rather than sketched: (a) class function; (b) Bernstein shape functions at $n = 8$; (c) the weighted terms and their sum for a fitted NACA 2412 upper surface; (d) the resulting surface against the section it was fitted to.

The property this paper exploits is that Eq. (4) is *linear* in A : the partial derivative

$$\frac{\partial \zeta}{\partial A_i} = C(\psi) S_i(\psi) \quad (5)$$

does not depend on A . It is a fixed matrix, computed once from the chordwise node distribution and cached. Recomputing it inside an iteration is the single most common way to lose the performance argument, and it is unnecessary. This design-independence is what makes the constraint block of Section 2.2 and the sensitivity matrix of the linear pre-solve in Section 2.7 constant operators assembled once rather than geometry-dependent operators re-derived at every step. It does *not*, by itself, make the flow-residual block $\partial R/\partial A$ cheap; that block requires the flow solver's own geometric sensitivity, which Section 2.4 shows is not available in complete form and must be obtained by finite differences.

Two endpoint identities follow directly from Eq. (4) and are used as constraint rows in Section 2.2. At $N_2 = 1$ the class function satisfies $C(1) = 0$ and $C'(1) = -1$, so $\zeta'(1) = \zeta_T - A_n$. This does *not* take the same form on both surfaces: with the boat-tail half-angle defined per surface as the wedge closing angle, the upper surface descends toward the trailing edge ($\tan \beta_u = -\zeta'_u(1)$) while the lower surface rises ($\tan \beta_l = +\zeta'_l(1)$), so

$$A_{u,0} = \sqrt{2R_{LE}/c}, \quad A_{l,0} = -\sqrt{2R_{LE}/c}, \quad A_{u,n} = \zeta_{T,u} + \tan \beta_u, \quad A_{l,n} = \zeta_{T,l} - \tan \beta_l. \quad (6)$$

The sign flip on the lower surface is stated explicitly because applying the upper-surface form to a lower-surface coefficient, which is negative under the stored-sign convention, silently produces the wrong constraint row; that error occurred once during development and is now pinned by a test against the fitted trailing-edge slope.

The leading-edge identity has a short derivation worth stating, because the curvature claim that follows from it is otherwise easy to misread. Near $\psi \rightarrow 0$ the class function gives $\zeta \approx A_0\sqrt{\psi}$, so in physical coordinates $z \approx A_0\sqrt{xc}$, and comparing with the osculating parabola $z^2 = 2R_{LE}x$ gives $R_{LE}/c = A_0^2/2$, the inverse of Eq. (6)'s first identity. Inverting the same expansion gives $x = z^2/A_0^2 + O(z^4)$, so the limiting curvature at the nose is $2/A_0^2$ and depends on A_0 alone; the next coefficient A_1 first enters at $O(z^3)$ and therefore affects the curvature *derivative*, not the curvature. Curvature continuity across the leading edge, G^2 continuity, consequently reduces to matching the leading coefficient magnitude on the two surfaces, $|A_{u,0}| = |A_{l,0}|$. Under the stored-sign convention ($A_{l,i} < 0$) the constraint row used to impose it is the stronger same-magnitude, opposite-sign statement $A_{u,0} + A_{l,0} = 0$, which also excludes the geometrically meaningless same-sign branch.

Figure 2 gives these identities their physical reading and checks the design-independence claim numerically. Panels (a) and (b) plot the two endpoint identities of Eq. (6) with the fitted NACA 2412 values marked, so the coefficients that the inverse solve returns can be read directly as a nose radius and a trailing-edge wedge angle rather than as abstract weights. Panel (c) is the one that matters for the Newton system: the markers are the surface displacement measured by actually perturbing one coefficient of the fitted section and re-evaluating the geometry, and the lines are the analytic prediction $\delta A_i C S_i$. They agree to the level printed on the panel, and the agreement does not depend on where the remaining coefficients sit, which is exactly what permits $\partial \zeta/\partial A$ to be assembled once and cached.

2.2 Constraints as linear rows

Because Eq. (4) is linear in A , every geometric functional that is itself linear or integral in ζ becomes a linear row $g \cdot A = b$ rather than a nonlinear predicate evaluated on a rebuilt mesh. The leading-edge-radius and trailing-edge-wedge rows follow immediately from Eq. (6). The one requiring a closed-form integral is inscribed area: the per-surface contribution is

$$\int_0^1 C(\psi) S_i(\psi) d\psi = K_i \int_0^1 \psi^{i+N_1} (1-\psi)^{n-i+N_2} d\psi = K_i B(i+N_1+1, n-i+N_2+1), \quad (7)$$

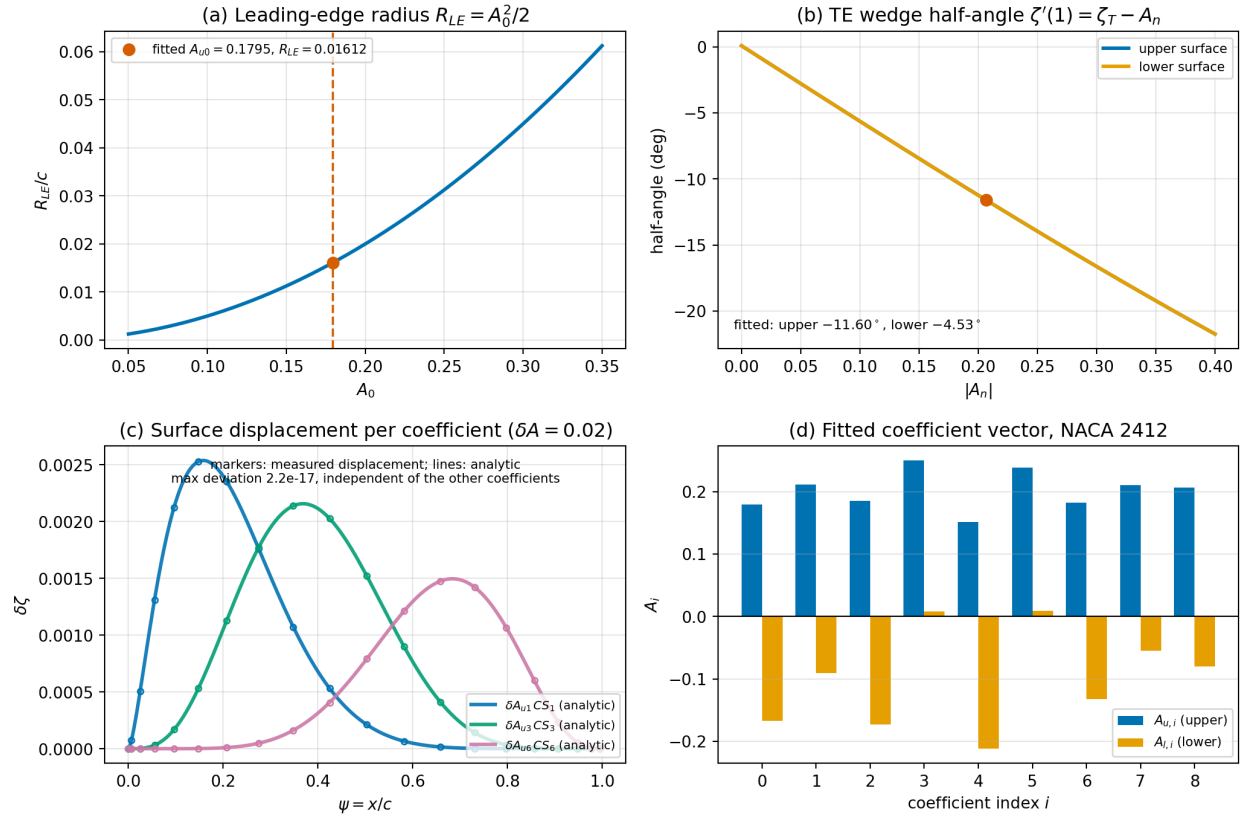


Figure 2: What the coefficients mean, and why the geometric sensitivity is constant: (a) leading-edge radius against A_0 ; (b) trailing-edge wedge half-angle against the final coefficient, both surfaces; (c) superposition measured against its analytic prediction; (d) the fitted coefficient vector in the ordering used throughout.

a Beta function, plus the trailing-edge term $\int_0^1 \psi \zeta_T d\psi = \zeta_T/2$. Appendix A supplies the derivation and the two-surface assembly, including the sign the lower block carries under the stored-sign convention. Equation (7) is evaluated with a library Beta function, never by numerical quadrature in a production constraint row; quadrature is retained only as an independent check, which the implementation passes, matching quadrature of the fitted NACA 2412 to 10^{-10} . This closed-form machinery is the same Beta-function apparatus Joseph and Mohan [2021] use for thin-airfoil lift and moment integrals against a Bernstein camber basis. This is not a coincidence: both integrate Bernstein polynomials against the class function, and it is the analytic kernel reused for the linear pre-solve of Section 2.7.

Table 1 sets these rows against what the same constraints require under a B-spline control-point wrapper, where every one of them becomes a nonlinear or non-smooth predicate evaluated on a rebuilt mesh. That is the stack that forced surrogate-based search in the author’s earlier turbine-blading work [Sathish et al., 2019], and the contrast is the reason the present formulation can dispense with an outer loop at all.

Table 1: Constraint representation under CST against a B-spline control-point wrapper.

Constraint	Under CST	Under the B-spline wrapper
Leading-edge radius	$A_{u,0} = \sqrt{2R_{LE}/c}$: linear, exact	implicit, via tangent points
Trailing-edge thickness	ζ_T term: linear	geometric post-check
Boat-tail wedge angle	A_n : linear	nonlinear angular design variable
Inscribed area	linear functional, Beta-function closed form	numerical, non-smooth
Curvature	rational function of A , differentiable	monotonicity predicate, non-smooth

Which rows the reported results actually exercise. The constraint machinery described here is broader than what the experiments in Section 5 use, and the difference should be stated rather than left to be inferred. Every configuration reported in this paper runs either with the leading edge prescribed, in which case $K = 0$ and no constraint row is assembled at all, or with the leading edge left free, in which case the single shared-leading-edge row $A_{u,0} + A_{l,0} = 0$ is used. The trailing-edge-wedge and inscribed-area rows, including the Beta-function derivation of Appendix A, are implemented and unit-tested against independent quadrature but are not exercised by any result reported here. They are presented as part of the formulation because they follow from the same linearity and cost nothing to assemble, not as validated contributions to the measured outcomes. Section 7.3 records the consequence: wherever the exactness of the constraint-row Jacobian is offered as an advantage over the nested baseline, it is an advantage of the formulation and not one this study has measured.

2.3 The extended Newton system and DOF accounting

The extended system appends the CST coefficients, and optionally the angle of attack, directly to the flow solver’s own global unknown vector. The unknowns are the flow state U of dimension N_{tot} , the CST coefficients A of dimension n_A , and the angle of attack α , contributing $n_\alpha \in \{0, 1\}$ unknowns. The equations are the flow residual $R(U; x(A)) = 0$, of which there are N_{tot} ; the target-matching rows $T(U) - Cp_{target} = 0$, of which there are M ; and the geometric constraint rows $G \cdot A - b = 0$, of which there are K .

Figure 3 shows the resulting bordered block structure, and Eq. (8) states it algebraically with the dimensions annotated for the reference configuration of Section 4. There, `mfoil`’s global system

carries $N_{sys} = 230$ stations, being 200 airfoil-surface nodes plus a 30-station wake, with four unknowns at each: the three boundary-layer states θ , δ^* and the amplification or shear-lag variable, plus the edge velocity u_e that closes the viscous-inviscid coupling. Hence $N_{tot} = 4N_{sys} = 920$. The prescribed-leading-edge treatment leaves $n_{A,free} = 16$ of the 18 CST coefficients as unknowns, and $M = 16$, $K = 0$, $n_\alpha = 0$. The assembled system is therefore $J \in \mathbb{R}^{936 \times 936}$, and the recorded rank at the first iteration is 936, that is, full:

$$J = \begin{bmatrix} \underbrace{\partial R / \partial U}_{N_{tot} \times N_{tot}} & \underbrace{\partial R / \partial A}_{N_{tot} \times n_{A,free}} & \underbrace{\partial R / \partial \alpha}_{N_{tot} \times n_\alpha} \\ \underbrace{\partial T / \partial U}_{M \times N_{tot}} & \underbrace{0}_{M \times n_{A,free}} & \underbrace{0}_{M \times n_\alpha} \\ \underbrace{0}_{K \times N_{tot}} & \underbrace{\partial G / \partial A}_{K \times n_{A,free}} & \underbrace{0}_{K \times n_\alpha} \end{bmatrix} \in \mathbb{R}^{(N_{tot}+M+K) \times (N_{tot}+n_{A,free}+n_\alpha)}, \quad (8)$$

square by Eq. (9)’s construction-time assertion ($M + K = n_{A,free} + n_\alpha$), so the $M + K$ extra rows exactly balance the $n_{A,free} + n_\alpha$ extra columns beyond the $N_{tot} \times N_{tot}$ flow block, which is itself always square.

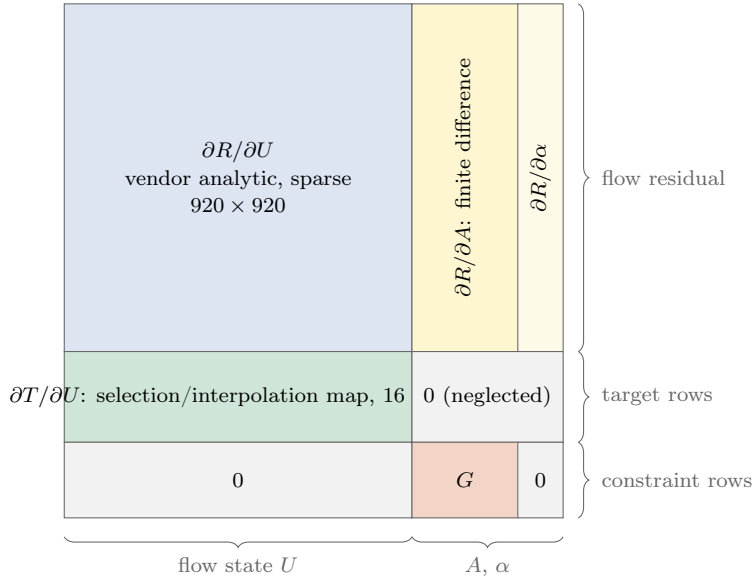


Figure 3: Block structure of the extended Jacobian at the reference configuration. The bordering rows and columns are the entire modification to the host solver’s own square system.

Three of the blocks are inherited unchanged and one is new. $\partial R / \partial U$ and $\partial R / \partial \alpha$ are `mfoil`’s own analytic, sparsely-assembled blocks, reused verbatim. $\partial T / \partial U$ maps the flow state to pressure coefficient at the target stations: a selection matrix when a station coincides with a panel node, and a two-node interpolation otherwise. $\partial G / \partial A = G$ is constant and free, per Section 2.2. The new block is $\partial R / \partial A$, which carries the architectural claim and whose construction is the subject of Section 2.4.

The zero block in the target rows is an approximation, not an identity. Under the original node-index addressing, in which each target row read the pressure coefficient at a fixed panel index, T depended on A only through U and $\partial T / \partial A$ was structurally zero. Target stations

are now addressed by surface and chordwise fraction, and the pressure coefficient at such a station is interpolated between the two bracketing nodes, $T_k = (1 - w)Cp_j + w Cp_{j+1}$. The weight w and the bracketing pair depend on node positions, which move with A , so $\partial T/\partial A = (\partial w/\partial A)(Cp_{j+1} - Cp_j)$ is small but not zero. The implementation neglects it. The neglect is defensible, because a coefficient perturbation moves the surface in z at a fixed chordwise grid and therefore perturbs w only weakly, and it is worth stating plainly because it has a measurable consequence: the iteration is an inexact Newton method, and Section 5.8 shows that the run using interpolated addressing converges at a fast but roughly constant rate rather than with the quadratic tail the node-indexed runs display. The same applies to the $\partial T/\partial \alpha$ block, which the implementation likewise assembles as zero on the grounds that pressure responds to incidence through the flow state; it is shown as a zero block in Eq. (8) for that reason.

Degree-of-freedom accounting. The original bookkeeping for this architecture counted rigid-body freedoms explicitly,

$$\text{DOF} = 2(n + 1) - 1_{\text{shared LE}} - 1_{\text{TE fixed}} + 3_{\text{scale/rotate/translate}} + 1_{\text{stagger or } \alpha} \stackrel{!}{=} M + K.$$

In the isolated-airfoil implementation reported here, the three rigid overall modes are not exposed as free Newton unknowns, so the implemented squareness condition is the simpler, equivalent statement actually asserted at construction time and logged on every run:

$$M + K = n_{A,\text{free}} + n_{\alpha}, \quad (9)$$

where $n_{A,\text{free}}$ is the count of CST coefficients *not* eliminated by a prescribed-leading-edge treatment or by an equality-row elimination, and $n_{\alpha} \in \{0, 1\}$ depending on whether angle of attack is a free unknown for the run in question. Appendix B works this accounting through for the reference configuration and states the general rule it specialises. Violating Eq. (9) raises at construction, before any Newton step. This is the guard against the rank and degrees-of-freedom failure mode of Section 3, and it is exercised directly in the ablation matrix (Section 5.1): deliberately over- and under-squaring the baseline system, at $n = 8$ per side with $n_{A,\text{free}} = 16$ and $n_{\alpha} = 0$, produces the exact, actionable diagnostics

```
"extended system not square: M+K=17 but
  n_A_free+n_alpha=16 (M=17, K=0, n_A_free=16)"
"extended system not square: M+K=15 but
  n_A_free+n_alpha=16 (M=15, K=0, n_A_free=16)"
```

with zero Newton iterations spent and no silent wrong answer. This matters because a rank-deficient extended system does not announce itself: it presents as Newton divergence, which is easily and wrongly attributed to the leading-edge singularity instead. Instrumenting for it directly, rather than inferring it from Newton behaviour, is the point of Eq. (9) being an assertion rather than a derived quantity.

2.4 Derivative strategy

The derivative-mode decision tree adopted for this work prefers, in order, the analytic chain rule $\partial R/\partial A = (\partial R/\partial x)(\partial x/\partial A)$ if the host solver exposes $\partial R/\partial x$ analytically; complex-step differentiation over A if the residual-assembly path is free of absolute value, minimum, maximum and comparison operators; and real central finite differences over A otherwise. Introspection of the host solver resolved both questions empirically, and neither resolved in the analytic chain rule's favour.

The analytic chain rule is incomplete for this solver class. `mfoil` exposes a residual sensitivity with respect to the surface arc-length mapping, but it is only a *partial* geometric sensitivity: real and useful for the arc-length coordinate, and in fact the route by which the solver carries its own stagnation-point sensitivity, but missing the inviscid panel and influence-coefficient coupling and the mass-influence operator through which a geometry change reaches the boundary layer. Using this partial block as if it were the complete $\partial R/\partial x$ would silently drop exactly the terms the leading-edge failure mode of Section 3 is about.

Complex-step differentiation is blocked, not merely inconvenient. A bare station-residual or transition-residual call, with fixed panels and a complex-perturbed *state*, is verified complex-step-safe and reproduces the corresponding analytic state Jacobian to double-precision machine epsilon. The *full* augmented residual, which is what a perturbation in A actually varies, since A moves the panel geometry itself, calls the panel-influence and stagnation-detection code paths. These use two-argument arctangent, branch selection on real quantities, and a hard type guard against complex node arrays. These are exactly the operators complex-step cannot tolerate, and they sit in precisely the geometry-dependent block that the analytic path was missing: complex-step cannot fill the gap the analytic chain rule leaves.

The implemented strategy is therefore real central finite differences with respect to A only, never with respect to node coordinates x . This is a deliberate architectural discipline, not a fallback of convenience. Over the Bernstein orders swept here, A carries between 8 and 24 free columns, so a full central-difference block costs $O(n_A)$ residual *evaluations* per Newton step, not flow solves, against the alternative of finite-differencing over $N_{panel} \gg n_A$ node coordinates, which would silently destroy the cost argument Section 5.7 depends on. The resulting Jacobian columns were cross-checked against the partial analytic block on the rows that block covers, agreeing there to five or six significant figures, and against a step-size sweep; no complete independent reference exists against which they could be verified in full. The step size itself, 10^{-6} , is not hand-tuned per case: the sweep varied it across several decades and found the resulting column agreement flat across the central range, with no visible truncation-against-roundoff cliff at the scales exercised, so 10^{-6} sits well inside a broad plateau rather than at a knife-edge optimum a different problem instance would move. Section 5.8 returns to the price of this choice, which is a residual floor near 10^{-10} that no Newton step can descend below.

2.5 Algorithmic summary

Figure 4 gives the whole procedure in one view before the pseudocode states it precisely. The two shaded regions are the paper’s two structural claims: everything left of the Newton loop is set up once and never revisited, and the loop itself solves a single square system rather than driving an outer optimiser. The two decision points are the only places the procedure can refuse to proceed, and both are diagnostics rather than tolerances: the squareness assertion of Eq. (9) is a statement about degrees of freedom, and the realisability gate is a statement about whether the target lies in the reachable set at all.

Algorithm 1 states the full monolithic iteration, target generation through release-and-verify, as pseudocode, making explicit where each of Sections 2.3 to 2.7’s mechanisms sits inside the loop: finite-difference Jacobian assembly, joint under-relaxation, the trust region on the coefficient step, and the forced-transition closure. It is a direct rendering of the implementation, not an idealisation of it. Algorithm 2 states the QR-pivoted station-selection procedure identified as the identifiability guard in Section 5.1, reusing the pre-solve sensitivity matrix rather than building a second one.

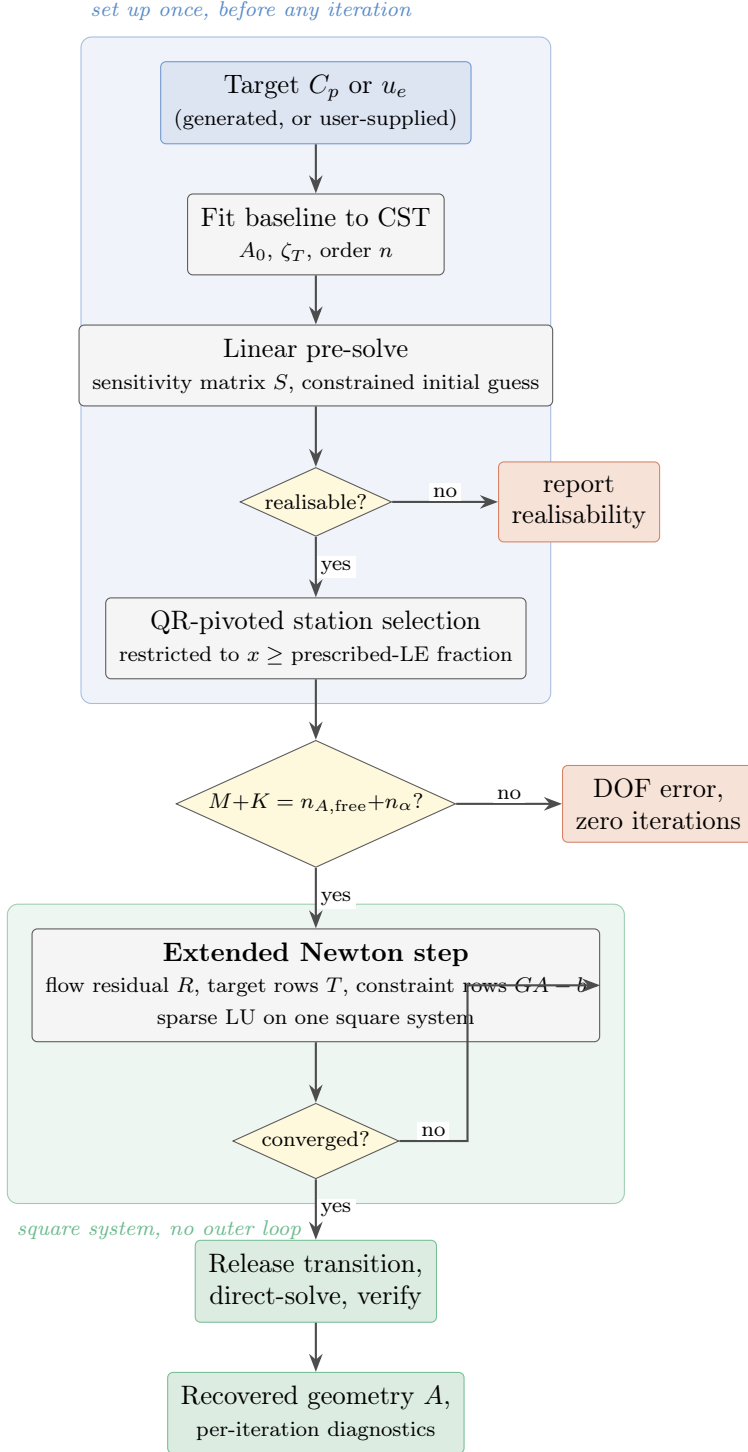


Figure 4: The monolithic inverse-design procedure. Pre-solve, station selection and the degree-of-freedom check happen once; the loop then solves one square system per step with no outer optimiser.

Algorithm 1 Monolithic CST–Newton inverse iteration

Require: target airfoil (for A^* generation) or externally supplied Cp_{target} ; Bernstein order n ; leading-edge treatment; transition mode

- 1: Fit CST to reference geometry $\rightarrow A^*$ (Section 2.1); build $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ once, cache (Eq. (5))
- 2: Generate target: direct-solve at A^* , with transition forced if the run so specifies (Section 2.6) $\rightarrow Cp_{target}$
- 3: Perturb or randomise $A^* \rightarrow A_0$; assemble G, b from Section 2.2
- 4: **if** initialisation is by pre-solve **then**
- 5: **for** 2 passes **do** ▷ Section 2.7
- 6: Build sensitivity matrix S at current A_0
- 7: Solve the constrained system Eq. (10) for A_0 ; record realisability and model gap as two *distinct* metrics
- 8: **end for**
- 9: **end if**
- 10: Select M target stations by QR pivoting (Algorithm 2), or evenly-spaced when ablating
- 11: **Assert** $M + K = n_{A,free} + n_\alpha$ (Eq. (9)) ▷ raises before any Newton step if violated
- 12: Direct-solve flow at A_0 ▷ initial point for the extended Newton loop
- 13: $k \leftarrow 0$
- 14: **while** $\|R\|, \|T\|, \|G\| > \text{tolerance}$ and $k < \text{iteration budget}$ **do** ▷ per-block residual norms recorded
- 15: Assemble $\partial R/\partial U, \partial R/\partial \alpha$ from the host solver (analytic, unchanged)
- 16: Assemble $\partial R/\partial A$ by central finite differences over A only (Section 2.4), $O(n_A)$ residual evaluations
- 17: Assemble $\partial T/\partial U$ (pressure-selection map), $\partial G/\partial A = G$ (constant)
- 18: Solve the linearised system $J\delta \leftarrow -[R; T - Cp_{target}; G \cdot A - b]$ (Eq. (8))
- 19: Compute joint under-relaxation $\omega \in (0, 1]$: the smaller of the host solver’s own state limiter and the coefficient trust region $\|\delta_A\|_\infty \leq \text{trust radius}$
- 20: $U \leftarrow U + \omega \delta_U$; $A \leftarrow A + \omega \delta_A$; $\alpha \leftarrow \alpha + \omega \delta_\alpha$ (if free)
- 21: Rebuild geometry from updated A ; record diagnostics
- 22: $k \leftarrow k + 1$
- 23: **end while**
- 24: **Release-and-verify** (Section 2.6): un-force transition; direct-solve at the converged A ; compare c_l, c_d against the reference section’s natural-transition values
- 25: **return** A , iteration count k , residual history, release-and-verify record

Algorithm 2 QR-pivoted target-station selection

Require: sensitivity matrix $S \in \mathbb{R}^{N_{cand} \times n_A}$ from the pre-solve (Section 2.7); free-coefficient index set $\mathcal{F}$; required station count $M_{req} = n_{A,free} - K + n_\alpha$

- 1: Restrict candidates to nodes at or aft of the prescribed-leading-edge fraction (Section 5.2)
- 2: Form the candidate submap $\tilde{S} \leftarrow S[\text{candidates}, \mathcal{F}]$
- 3: Column-pivoted QR of $\tilde{S}^\top$: $\tilde{S}^\top P = QR$, whose pivot ordering ranks candidate stations by decreasing residual norm in the orthogonal complement of those already chosen
- 4: Select the first M_{req} pivoted candidate indices $\rightarrow$ target stations
- 5: **return** sorted station index set; report $\text{cond}(\tilde{S}[\text{stations}, :])$, a quantity *distinct* from $\text{cond}(J)$

2.6 Forced-transition onset closure

The e^n transition closure used by `mfoil`, and by XFOIL before it, is not smooth in the design variables. Within a station the internal solve for the point at which the marched amplification factor crosses n_{crit} is smooth, but the surrounding bookkeeping is not: as geometry changes, the index of the last laminar node relocates discretely, and that relocation re-initialises the shear-lag state across the newly turbulent nodes through the transition correlation. A coupled Newton iteration needs a continuous residual; a jump of this kind stalls or chatters it. The mitigation is to fix, or trip, transition during the inverse iterations and release it for a verification solve afterward. `mfoil` exposes no forced-trip interface, so this is implemented as an adapter-level replacement rather than a vendor-code edit, in keeping with the discipline that the vendored solver is never modified.

The first implementation attempt froze the laminar and turbulent node flags and suppressed the solver’s own transition update, but left the natural transition-station equation untouched at the frozen boundary station. This was *insufficient*. Empirically, on NACA 2412 at $\alpha = 2^\circ$ and $Re = 10^6$ with transition tripped at 5 percent chord on both surfaces, the physics moved correctly under Newton pressure, with c_d rising from 0.00578 to 0.00741, but the solve never converged: the residual norm stalled near 12.6 for the full 50-iteration budget, because the frozen station equation still solves internally for amplification equal to n_{crit} , which has no solution once transition is pinned somewhere amplification is nowhere near n_{crit} .

The corrected mechanism replaces the boundary-station residual with an XFOIL-style turbulence-*onset* closure. The momentum and shape-parameter rows use the ordinary turbulent station closure directly across the whole station, with no laminar sub-leg and no station-local transition point, and the shear-lag row is replaced with the same onset condition the solver’s own transition update uses to initialise a newly-turbulent node, namely that the shear-lag state equals the transition correlation evaluated at the downstream node. Because that correlation is a function of the downstream state alone, the replacement row has a particularly simple Jacobian structure, stated in Appendix C. With this closure the trip converges cleanly: at 5 percent chord the solver reports convergence with residual norm below 10^{-10} and $c_d = 0.01125$, strictly greater than the natural-transition $c_d = 0.005778$, as expected for the larger turbulent wetted area. A trip at 30 percent gives $c_d = 0.00788$, between the 5 percent and natural values and consistent with the expected decrease of drag as the trip moves aft, though three points spanning two different residual closures do not by themselves establish monotonicity. Releasing transition and re-solving from the converged forced state reproduces the pinned natural baseline within its stated tolerance, $c_l = 0.449351 \pm 10^{-3}$ and $c_d = 0.005778 \pm 2 \times 10^{-4}$. These figures come from the closure’s own verification run and differ from the release-and-verify values quoted in Section 4 ($c_l = 0.449091$, $c_d = 0.005774$) in the fourth and sixth decimal respectively; that is a different run at a different paneling, not a restatement of the same one. Rebuilding the global system after convergence with the replacement closure still active reproduces a residual below 10^{-8} : the converged state is a genuine fixed point of the forced residual, not an artifact of the step sequence. Every result reported in Section 5 therefore states its transition mode explicitly, and release-and-verify in natural-transition direct mode is part of the protocol (Section 4), not an optional extra.

Figure 5 shows the state this closure operates on, and is the reason a coupled solver was required rather than an inviscid one with a boundary-layer post-process. The inverse system of Section 2.3 is appended to `mfoil`’s *viscous* global system, so the six distributions in the figure are unknowns inside the same Newton solve that recovers the geometry, not quantities computed afterwards from a converged pressure field. The coupling itself is a mass defect $m = u_e \delta^*$ entering the inviscid problem as a transpiration source through the solver’s mass-influence operator, rather than a literally displaced body; panel (f) plots the displacement surface $\zeta \pm \delta^*$ as the conventional visualisation of

that coupling. A geometry perturbation reaches the boundary layer principally through the inviscid edge velocity and the arc-length metric, and returns through the mass defect. The trip locations are marked on each panel; the skin-friction rise at the trip in panel (c) and the corresponding drop in shape factor in panel (d) are the closure described above taking effect.

NACA 2412, $\alpha = 2.0^\circ$, $Re = 1.00e + 06$, transition forced at $x/c = 0.05$ upper / 0.05 lower; $c_l = 0.4417$, $c_d = 0.01125$

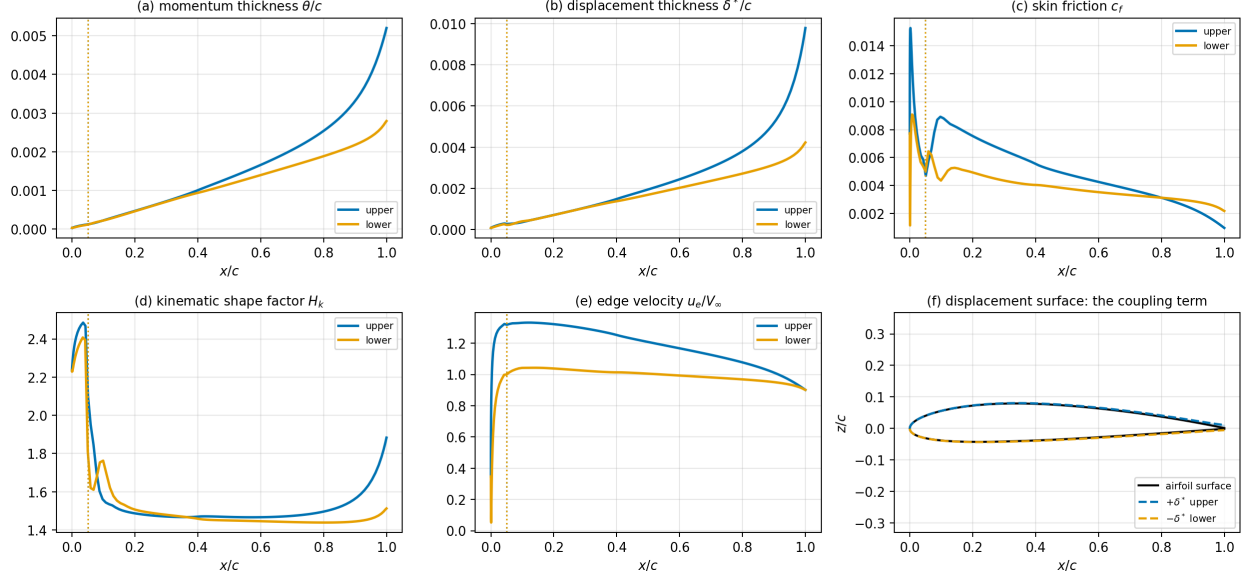


Figure 5: The coupled viscous solution the inverse system is embedded in, for NACA 2412 at $\alpha = 2^\circ$, $Re = 10^6$, transition forced at 5 percent chord (dotted): (a)–(b) momentum and displacement thickness; (c) skin friction; (d) kinematic shape factor; (e) edge velocity; (f) the displacement surface.

2.7 Analytic pre-solve and realisability

Two jobs are done by a single linear pre-solve before any Newton iteration runs, and the distinction between them is load-bearing for how results are reported. Thin-airfoil theory is linear in camber slope and thickness, and CST camber and thickness are linear in A , so the pressure response to a coefficient increment is well approximated by a sensitivity matrix S built either analytically, from the Beta-function kernels of Joseph and Mohan [2021], or, as implemented, numerically from `mfoil`'s own linear-vorticity panel influence coefficients at fixed geometry. Solving the constrained least-squares problem with the Section 2.2 rows as linear equality constraints,

$$\begin{bmatrix} S^\top S & G^\top \\ G & 0 \end{bmatrix} \begin{bmatrix} \delta A \\ \lambda \end{bmatrix} = \begin{bmatrix} S^\top (Cp_{target} - Cp_0) \\ b \end{bmatrix}, \quad (10)$$

where Cp_0 is the pressure distribution at the current coefficient vector, serves two purposes. It gives an initial guess placing Newton inside its basin of attraction, which matters because the leading-edge region is exactly where Newton wanders and the host solver's own documentation warns that the coupled solve can fail from a poor initial guess. It also gives a *realisability* metric, $\|S \delta A - (Cp_{target} - Cp_0)\| / \|Cp_{target}\|$, that flags targets outside the CST-representable manifold *before* a Newton solve is attempted. The increment form matters: it is computable for a target whose generating coefficients are unknown, which the difference against a notional true A^* would not be.

An adversarial review of the pre-solve found these two jobs’ outputs had been conflated into one number, and the resulting reporting convention is binding for every number in this paper. First, CST-representability, the realisability guard above, is answered with *model-consistent* quantities: an inviscid target against an inviscid linearisation. On the reference configuration this is 3.90×10^{-4} , and the target is representable. Second, the *model gap*, how far the inviscid linearisation sits from the viscous physics the Newton solve actually faces, is a separate, diagnostic-only quantity, 0.110 on the same case. That is boundary-layer displacement effect, which no inviscid model captures. It is *not* target unrealisability; the subsequent monolithic solve converging to 2.75×10^{-11} (Section 4) confirms, for this self-generated target, that the target was realisable despite the nonzero model gap.

Two reporting rules follow and are enforced throughout. Realisability numbers are always labelled with the model they were computed under. And the conditioning of the QR-selected station submap, approximately 90 at the reference configuration (Section 5.1), is never quoted as the conditioning of the full extended Newton system. These are different matrices answering different questions, and conflating them was one of the review’s confirmed findings.

3 Failure modes and their instrumentation

Four failure modes were identified before implementation as the ways the architecture of Section 2 can fail without that failure being obviously attributable to its true cause. They are listed in Table 2 with the diagnostic that detects each and the mitigation adopted. The list is deliberately structured around *discriminability* rather than around severity: a coupled Newton solve that stalls looks the same whatever caused it to stall, so the instrumentation exists so that a failed solve reports which mode it hit rather than only that it failed.

Table 2: The four failure modes identified before implementation, with the diagnostic that detects each and the mitigation adopted.

Failure mode	Mechanism	Diagnostic	Mitigation
Rank and degrees of freedom	extended system rank-deficient or over-determined	rank and condition number of J	squareness assertion, Eq. (9); realisability pre-solve
Basis conditioning	Bernstein Gram matrix conditions like 4^n	Gram condition number against n	keep $n \leq 10$ per side; orthogonalise the basis
Leading-edge row conditioning	$\sqrt{\psi}$ non-analytic at the nose inflates nearby residual rows	Jacobian row norm against station	row scaling; prescribed leading edge (Section 4)
Transition-closure non-smoothness	discrete relocation of the transition index kinks the residual	transition-station history	forced trip, then release and verify (Section 2.6)

Six per-iteration diagnostics support that discrimination, and Table 3 states what each is for. Two of them are not tied to a single failure mode. The per-block residual norms discriminate *which* block of Eq. (8) is stalling, which is the first question to ask of any non-convergence, and the Newton convergence history is the general-purpose instrument used throughout Section 5 and the basis of the convergence-rate analysis in Section 5.8. The remaining four map one-to-one onto the rows of Table 2.

One entry in Table 2 deserves expansion, because its mechanism is routinely misdescribed and the correct description is what determines the mitigation. The leading-edge failure mode targets *rows*, not columns. Section 2.1 showed that $\partial\zeta/\partial A_i = C(\psi)S_i(\psi)$ is perfectly finite as $\psi \rightarrow 0$, tending to

Table 3: Per-iteration diagnostics recorded on every run.

Diagnostic	Quantity recorded	Question it answers
Per-block residual norms	$\ R\ $, $\ T\ $, $\ GA - b\ $ separately	which block of the extended system is stalling
Rank and conditioning	rank and $\text{cond}(J)$ at each iteration	is the extended system square and full-rank in practice
Basis conditioning	condition number of the Bernstein Gram matrix, and of the selected station submap	is the parameterisation, or the sampling, the limiting factor
Row-norm profile	Jacobian row norm against chordwise station	are the leading-edge rows dominating the matrix
Transition history	transition index and location per iteration	is the transition closure moving discontinuously
Convergence history	residual norm against iteration	what rate is the iteration actually achieving

zero there, so the geometric columns are healthy. What blows up is the flow residual’s *sensitivity* to geometry near the nose: surface normals, metric terms and the stagnation-point mapping all have unbounded derivatives there, so a handful of rows near the stagnation point acquire enormous magnitude and destroy the coupled matrix’s condition number. This is a concrete instance of the point made in Section 2.1: CST’s linearity keeps the columns exact, but does not by itself guarantee the rows are well-conditioned near a genuinely non-analytic geometric feature. It is also why the mitigation is geometric, prescribing the nose rather than reweighting the basis, and why prescribing the nose in turn forces the station-restriction rule of Section 5.2: the two are a matched pair, and applying either alone reintroduces the degeneracy from the opposite direction.

All four modes were predicted before implementation, and three of the four were subsequently observed in the runs reported here: the rank mode in the deliberately mis-squared configurations of Section 5.1, the basis-conditioning mode in the order sweep of Section 5.3, and the transition-closure mode in the stalled first implementation described in Section 2.6. The leading-edge mode is the exception. Its mitigation was adopted from the outset, so it is inferred from the station-restriction experiment of Section 5.2 rather than observed directly in an unmitigated run; the row-norm diagnostic that would show it is recorded but not presented here.

4 The falsifiable test

The architecture’s central claim, that appending CST coefficients to `mfoil`’s Newton system yields a determined root-find rather than a search, is falsifiable by a single self-consistency experiment with a *known* answer, guaranteed realisable by construction. The protocol fits CST to a known airfoil, runs `mfoil` in direct mode to generate a target pressure and edge-velocity distribution from that fit, perturbs the coefficients or uses the pre-solve of Section 2.7 to obtain a starting guess, and asks whether the monolithic inverse Newton iteration recovers the original coefficient vector A^* , following the procedure stated as pseudocode in Section 2.5. Any failure of this test isolates cleanly to one of the Table 2 failure modes rather than to target infeasibility. This is why this test, and not a search over independently chosen targets, is the load-bearing evidence for the hypothesis. It is also, by the same construction, the test’s principal weakness, and Section 7.2 states that weakness in full rather than leaving it implicit here.

Reference configuration. NACA 2412, $n = 8$ per side, leading edge prescribed, meaning the first 5 percent of chord is fixed geometrically rather than left inverse. That is the mixed direct-inverse

mitigation for the leading-edge failure mode of Section 3, and it is the cheapest test of the whole hypothesis because it isolates the leading-edge question from the architecture question. Transition is forced during the inverse iterations (Section 2.6), target stations are chosen by QR column pivoting (Section 5.1), the iteration is initialised from the linear pre-solve, and the angle of attack is held fixed. This configuration is referred to throughout the rest of the paper as the reference configuration.

Protocol, including release-and-verify. The success criterion is $\|A - A^*\|_\infty < 10^{-4}$ within nine Newton iterations, together with a residual history showing sustained superlinear decrease. Because the inverse solve runs under the forced-transition closure of Section 2.6, the protocol additionally requires releasing transition and re-solving in fully natural, direct mode at the recovered geometry, then comparing the resulting aerodynamic coefficients against those of the *reference section* solved in the same natural mode. This is a check that the forced-transition inverse solve recovered the geometry, rather than merely finding a zero of the forced-mode residual that happens to be aerodynamically different.

Result. Measured against the 10^{-4} and nine-iteration criterion, the solve returns $\|A - A^*\|_\infty = 2.747 \times 10^{-11}$ in six Newton iterations, with the target-block residual history $5.85 \times 10^{-4} \rightarrow 2.05 \times 10^{-5} \rightarrow 1.57 \times 10^{-7} \rightarrow 1.21 \times 10^{-8} \rightarrow 2.13 \times 10^{-10} \rightarrow 9.47 \times 10^{-12}$, plotted alongside the ablation runs in Figure 16. Release-and-verify against a $\Delta c_l < 10^{-3}$, $\Delta c_d < 2 \times 10^{-4}$ criterion gives $\Delta c_l = 3.4 \times 10^{-12}$ and $\Delta c_d = 3.0 \times 10^{-14}$, both essentially at floating-point noise: the released solve at the recovered geometry reproduces the reference section’s natural-transition coefficients ($c_l = 0.449091$, $c_d = 0.005774$) to machine precision. Because both sides of that comparison are natural-mode solves at geometries agreeing to 2.7×10^{-11} , agreement at this level is what smoothness alone would predict; the check confirms that no aerodynamically distinct root was reached, and should not be read as independent evidence of accuracy. Realisability, computed inviscid-consistently as Section 2.7 requires, is 3.90×10^{-4} ; the model gap, diagnostic only, is 0.110. Sixteen coefficients are *recovered* by the Newton iteration; the two leading-edge coefficients $A_{u,0}$ and $A_{l,0}$ are *prescribed* by the configuration and are not recovered, so all error norms quoted here are over the 16 free coefficients. Table 4 collects the criteria and the measured values.

Table 4: The falsifiable test at the reference configuration: every criterion met, each by several orders of magnitude.

Criterion	Threshold	Measured
$\ A - A^*\ _\infty$ over the 16 free coefficients	$< 10^{-4}$	2.747×10^{-11}
Newton iterations	≤ 9	6
Sustained decrease of the target block	required	7 decades over 5 steps
Release-and-verify Δc_l	$< 10^{-3}$	3.4×10^{-12}
Release-and-verify Δc_d	$< 2 \times 10^{-4}$	3.0×10^{-14}

An independently-run six-vector adversarial review probed circularity, station-index correspondence, the pinned-coefficient framing, the absence at the time of a release-and-verify step, realisability semantics, and the conflation of submap conditioning with system conditioning. Every confirmed finding is reflected in this paper’s reporting discipline: the split between free and prescribed coefficients above, and the two-metric realisability rule of Section 2.7. The station-index correspondence finding was closed differently from the others. It was originally answered with a numeric guard, asserting that the selection-time node and the target’s own paneling agreed to $\max |\Delta x/c| < 2 \times 10^{-5}$.

That guard has since been removed, because addressing a station by surface and chordwise fraction, and interpolating the target’s own pressure curve at that fraction, makes the correspondence exact rather than approximate, so there is no longer a quantity for the guard to bound. Two consequences are recorded here rather than left implicit. Interpolated addressing is what introduces the neglected $\partial T/\partial A$ term of Section 2.3, and with it the inexact-Newton behaviour analysed in Section 5.8. And the runs of Section 5, which predate the change, retain the earlier nearest-node addressing with a residual correspondence error of up to 2.0×10^{-4} in chord fraction, ten times the bound the retired guard imposed; Section 5 states where that matters.

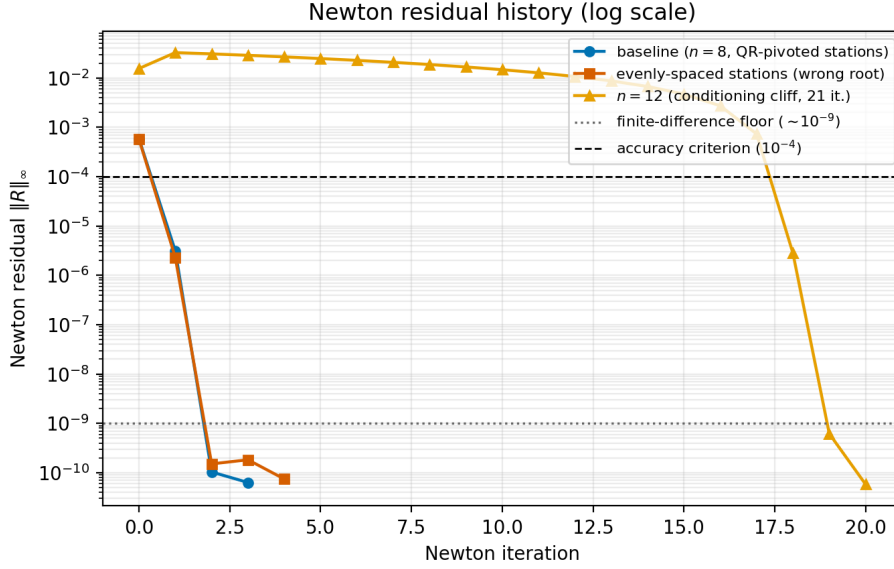


Figure 6: Newton residual history of the node-indexed baseline run, overlaid with the evenly-spaced-station wrong-root case and the $n = 12$ conditioning-cliff case of Section 5.1.

Figure 6 places that history in context by overlaying the node-indexed baseline against the two ablation runs that fail for different reasons, the evenly-spaced-station case that converges to a wrong root and the $n = 12$ case that converges slowly; all three are analysed in Section 5.1. A residual history is in any case a scalar summary. Figures 7 to 9 give the geometry and flow-field view of the same result, regenerated directly from a fresh re-run of the reference configuration rather than hand-plotted. Figure 7 overlays the target and Newton-recovered airfoil outlines, visually indistinguishable at full-chord scale, with a leading-edge inset showing the same at the scale where the leading-edge failure mode would be visible if the prescribed-nose mitigation were absent. It also includes a per-coefficient recovery-error bar chart making explicit which two of eighteen coefficients are pinned rather than recovered.

Figure 8 compares target and recovered pressure distributions both as a full curve and restricted to the 16 QR-pivoted stations the Newton system actually solved against. The distinction matters: agreement at the solved stations is enforced by construction, whereas agreement away from them is not, and it is the full curve that shows the recovered geometry reproduces the target everywhere rather than only where it was sampled. This is the pressure-level counterpart of the coefficient-level agreement in Table 4, and rules out the possibility that the reported error is an artifact of comparing only at the fitted stations.

Figure 9 shows the converged, released, natural-transition flow solution at the recovered geometry: the pressure distribution and the boundary-layer displacement thickness. This is the physical state

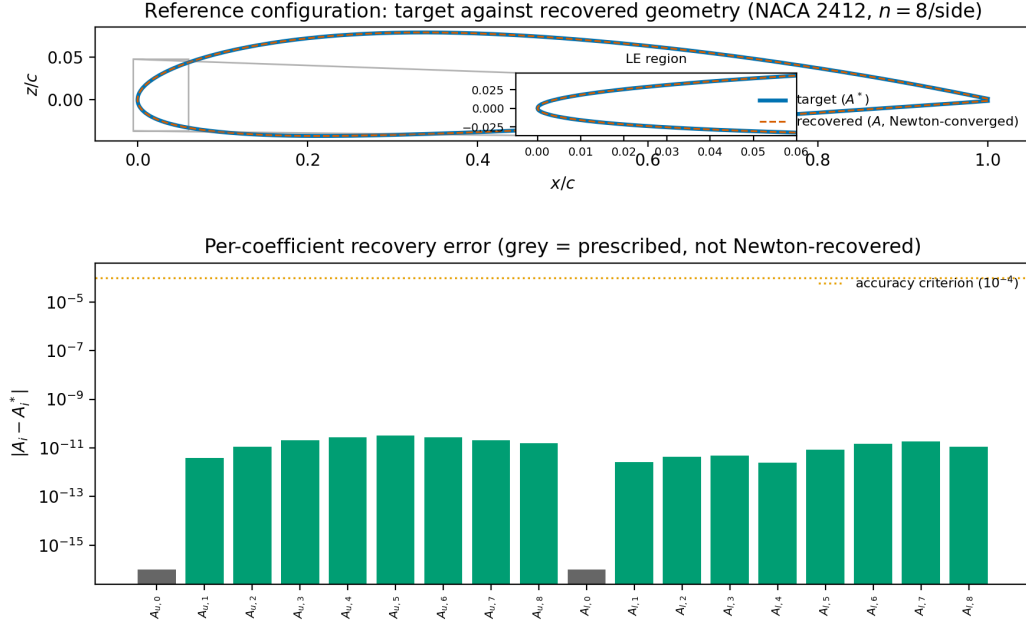


Figure 7: Target against Newton-recovered geometry at the reference configuration, with a leading-edge inset and a per-coefficient recovery-error bar chart. Grey bars are prescribed, not recovered.

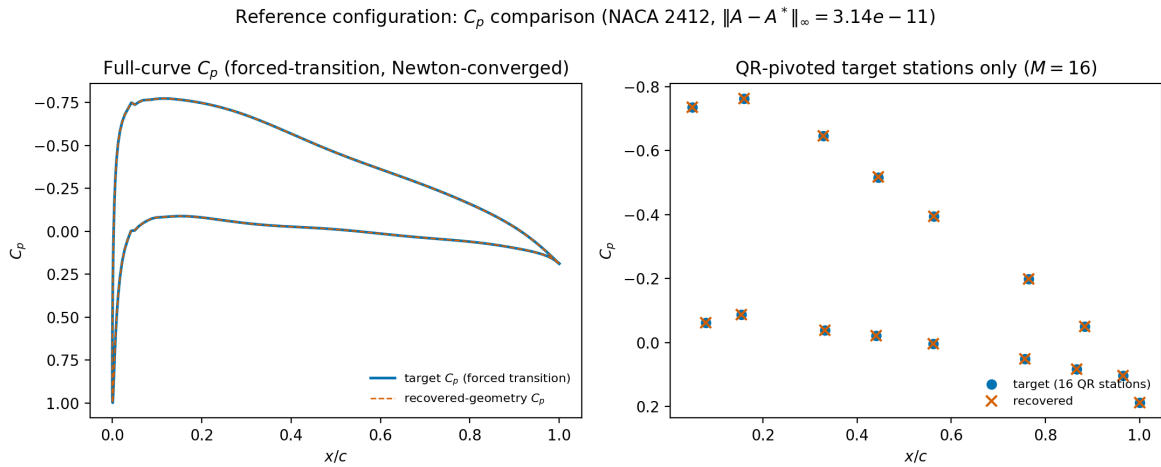


Figure 8: Target against recovered pressure distribution under forced transition, plotted in the aerodynamic convention with inverted vertical axis: full curve (left) and the 16 QR-pivoted target stations only (right).

from which the release-and-verify comparison of Table 4 was computed, and the displacement-thickness panel is worth reading alongside Figure 5: with transition released, the laminar run extends well aft of the 5 percent trip used during the inverse iterations, which is precisely the difference the release step exists to check the solve did not absorb into geometry.

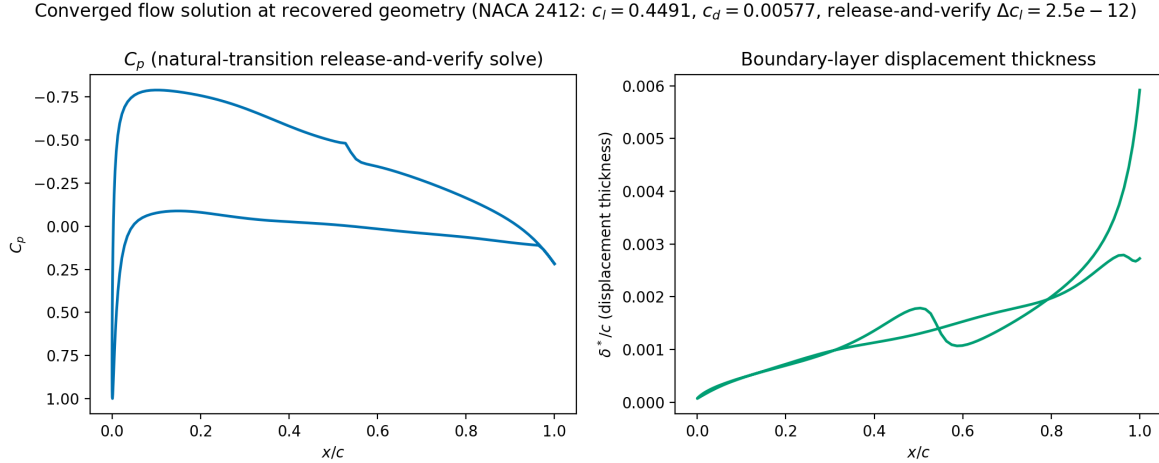


Figure 9: Converged flow solution at the recovered geometry after transition is released: pressure distribution (left) and displacement thickness δ^*/c (right).

5 Ablations and results

Every number in this section traces to an archived run record, and Appendix F names the record behind each table and figure. Unless otherwise stated, values are single runs at a fixed seed; no confidence intervals are reported for them because none is statistically defensible at one run per configuration. The order sweep, the station-selection comparison and the other single-factor *ablations* in this section are a single-airfoil matrix on NACA 2412 plus two additional single-airfoil runs on NACA 0012 and 4415. These are deliberately narrow comparisons designed to isolate one factor at a time, not panel studies, and convergence-fraction language is avoided in those subsections for that reason. Sections 5.5 and 5.6 are the exceptions: they report the two pre-registered panels, run subsequently at the configuration these ablations identify, and their interval estimates are stated there on their own terms.

Station-addressing provenance. The ablation matrix in this section was run with target stations addressed by panel-node index, which was the implementation at the time. Addressing was subsequently changed to surface and chordwise fraction with interpolation between the two bracketing nodes, because a node index stops denoting the same physical location once the geometry moves, which is precisely what an inverse solve does. The change moves the reference-configuration figures quoted in Section 4 from 1.08×10^{-11} in four iterations to 2.75×10^{-11} in six, both far inside the criterion, and leaves the QR-selected submap conditioning unchanged at 90.3. Every run in this section shares the earlier addressing, so the within-matrix comparisons the ablations rest on are unaffected; what must not be done is to read a single number from this section alongside a single number from Section 4 as though they came from one run. Where that distinction matters it is stated at the point of use.

5.1 Station-selection identifiability

This finding leads the ablation discussion, ahead of the cost comparison, because it is the most novel, most precisely quantified and most consequential result reported here. The square extended Newton system (Section 2.3) demands exactly M target-station equations. With evenly-spaced stations, between-station pressure information is discarded, and the resulting 16×16 station-sensitivity submap can be badly ill-conditioned. Table 5 reports the two configurations that isolate this mechanism.

Table 5: Station-selection ablation. Both runs report convergence and drive the residual to the solver floor, yet their recovery errors differ by eight orders of magnitude.

Station selection	$\ A - A^*\ _\infty$	Submap condition	Reported converged
QR-pivoted (reference configuration)	1.08×10^{-11}	90.3	yes
Evenly spaced	1.43×10^{-3} (misses the 10^{-4} criterion)	3.47×10^{12}	yes

Both runs’ Newton loops report their own internal convergence, and the residual reaches the solver’s floor in both cases. Only QR-pivoted selection recovers the correct design. Converged and correct are different claims, invisible unless the coefficient error is checked against the true coefficients rather than the residual norm alone. The mechanism is that with evenly-spaced stations there exist near-null coefficient directions whose motion changes the sampled pressure by less than floating-point precision. Newton then converges to *any* of several exact roots of the badly-conditioned discrete system: a structural non-uniqueness of the sampled problem, not a numerical accident. Two contributing null-direction families were identified and eliminated *before* the station-selection fix, each verified empirically. The first is a camber–incidence equivalence: freeing α lets a slightly different camber at a slightly different incidence interpolate the same samples, so α is held fixed for self-consistency tests. The second is a prescribed-nose mismatch: excluding leading-edge stations while leaving the leading-edge-dominant coefficients free leaves A_0 unobservable, which is why the prescribed treatment *fixes* $A_{u,0}$ and $A_{l,0}$ rather than merely dropping leading-edge target rows.

The resolution keeps the system square, with no least-squares relaxation and no reintroduced optimisation, and instead makes each of the M equations maximally informative. The pre-solve of Section 2.7 already builds the full $200 \times n_A$ sensitivity matrix for initialisation; reusing it to *select* the M target stations by QR column pivoting chooses rows greedily by decreasing residual norm in the orthogonal complement of those already chosen. That is a standard rank-revealing heuristic rather than a conditioning optimum, since optimal subset selection is combinatorial, but its measured effect here is decisive: submap condition 3.47×10^{12} against 90.3, and recovery error 1.43×10^{-3} against 1.08×10^{-11} . An earlier, uncontrolled observation of the same mechanism had reported a submap condition near 10^7 with recovery errors between 1.4×10^{-3} and 6×10^{-2} ; the controlled comparison here reproduces the effect and sharpens it, and the two conditioning figures should not be read as measurements of the same quantity. Full-curve sensitivity information enters only through the already-required linear pre-solve, and the nonlinear Newton solve stays square and determined throughout. Sensitivity-optimal station selection is therefore the identifiability guard for determined inverse systems, a methodological contribution beyond the original hypothesis: a practical answer to the non-uniqueness caveat inverse problems inherit from Lighthill’s and Volpe and Melnik’s well-posedness constraints (Section 1.1), without abandoning the root-finding architecture for a least-squares relaxation.

One asymmetry in this comparison should be recorded. The two arms do not draw from a bit-identical candidate set: the evenly-spaced arm additionally omits the two nodes flanking the stagnation point, and the two arms index slightly different chordwise grids. The effect is far too small to account for eight orders of magnitude in recovery error, and the conditioning gap is itself eleven orders, but the comparison is not a strictly single-factor one and is not presented as such.

Figure 10 summarises the ablation: the same solver, the same target and the same convergence flag produce recovery errors eight orders of magnitude apart, separated only by which stations were sampled.

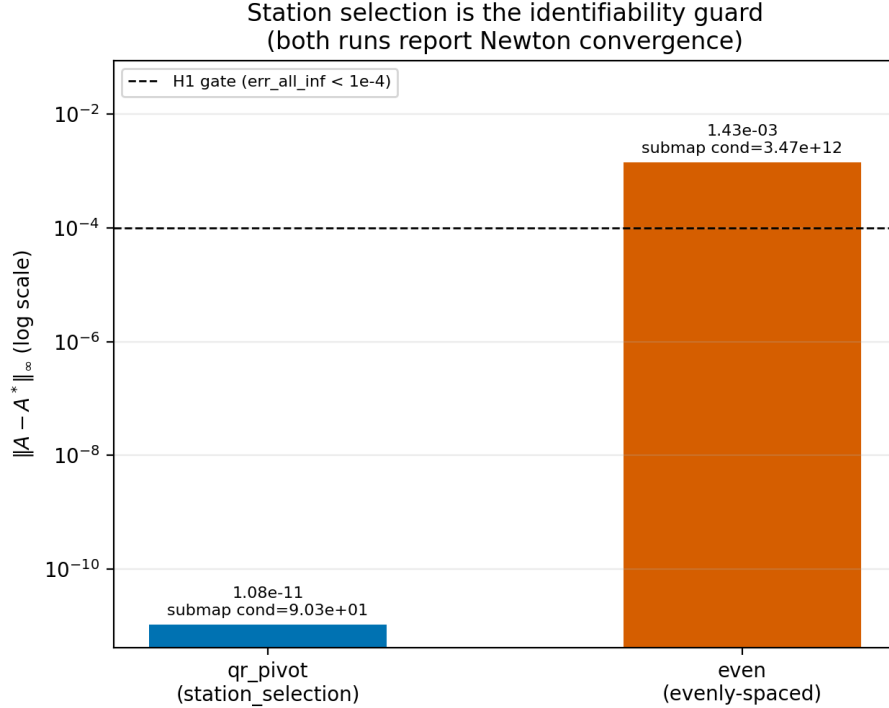


Figure 10: Recovery error under QR-pivoted against evenly-spaced target stations, annotated with the conditioning of each station submap.

5.2 Where the stations are placed: the prescribed leading edge

Section 5.1 records one half of a requirement: excluding leading-edge target stations while leaving the leading-edge-dominant coefficients free leaves A_0 unobservable, which is why the prescribed treatment fixes $A_{u,0}$ and $A_{l,0}$ rather than merely dropping leading-edge rows. The other half is the converse, and it was found while extending the method to arbitrary user-supplied targets: once the leading edge *is* prescribed, the leading-edge target stations must in turn be excluded. The two conditions are not independent choices. Prescribing the nose without restricting the candidate stations, or restricting the stations without prescribing the nose, each leaves a near-null direction in the sampled system; only the pair is identifiable.

Under the prescribed treatment, $A_{u,0}$ and $A_{l,0}$ are given rather than solved, so a target row placed inside the prescribed region constrains pressure over a piece of surface that cannot move. The effect on sampling is not a uniform loss of information, and Figure 11(a) is worth reading carefully on this point. Across the prescribed region the norm of the pressure sensitivity to the free

coefficients spans nearly two orders of magnitude, from 0.054 at the nose itself to 3.90 just aft of it, against a median of 1.47 over the rest of the chord. The region therefore contains both the single least informative station on the surface and some of the most sensitive. That combination is what makes unrestricted QR selection a poor choice here rather than merely a neutral one: the pivoting is drawn to the high-sensitivity nose stations, and the rows it selects there are strongly mutually correlated, because a small neighbourhood of a fixed nose produces nearly the same pressure response for every free coefficient. The extended system stays formally square and full-rank while becoming numerically near-dependent, which is the same underlying condition as Table 5 reached from the opposite direction.

Restricting the QR candidate set to stations at or aft of the prescribed-nose fraction, as Algorithm 2 specifies, removes the degeneracy. Table 6 reports the effect on a viscous self-consistency target, holding everything else fixed. The comparison is run at $n = 6$ per side, not the $n = 8$ of the ablation matrix above, so it is a controlled single-factor experiment in its own right rather than an additional entry in that matrix; its numbers should not be read against the $n = 8$ rows of Table 7. Like the rest of this section it is a single run per arm.

Table 6: Target-station placement under a prescribed leading edge, at Bernstein order $n = 6$ per side. Both arms start from the same seeded perturbed geometry. Surface offset is in millichords.

Quantity	Leading-edge stations in candidate set	Leading-edge stations excluded
Candidate stations	200	169
Submap condition	31.7	19.9
Converged	no	yes
Newton iterations	50 (budget exhausted)	7
Final residual	1.89×10^{-10}	8.21×10^{-11}
$\ A - A^*\ _\infty$ over free coefficients	6.97×10^{-1}	4.10×10^{-3}
Maximum surface offset (millichords)	11.74	0.127

The residual is not what separates the two columns. The unrestricted arm drives its residual to 1.89×10^{-10} , within an order of magnitude of the restricted arm’s, and still fails to converge inside a 50-iteration budget while ending 11.7 millichords from the target surface. Restricting the candidate set converges in seven iterations to 0.127 millichords, a ninetyfold improvement in the recovered shape, with the submap conditioning improving only from 31.7 to 19.9. A conditioning number that moves by less than a factor of two therefore does not bound the effect: what matters is whether the equations are spent on a part of the surface the solver is allowed to move. For scale, a CST fit of this order is itself permitted a root-mean-square deviation of one millichord from the section it represents, so the restricted arm’s peak offset sits inside the representation’s own tolerance while the unrestricted arm’s exceeds it by an order of magnitude.

This reinforces rather than repeats Section 5.1. There, the degeneracy came from discarding between-station information across the whole chord, and both configurations converged, one of them to a wrong root. Here it comes from spending equations on a region that has been removed from the design space, and the failure is visible as non-convergence rather than as a confident wrong answer. Neither is detectable from the residual norm alone.

5.3 Bernstein order: the basis-conditioning cliff

Table 7 sweeps Bernstein order n per side from 4 to 12, reporting the three distinct conditioning quantities alongside iteration count and recovery error. These are three different matrices, reported separately under the rule stated in Section 2.7: the condition number of the selected station submap,

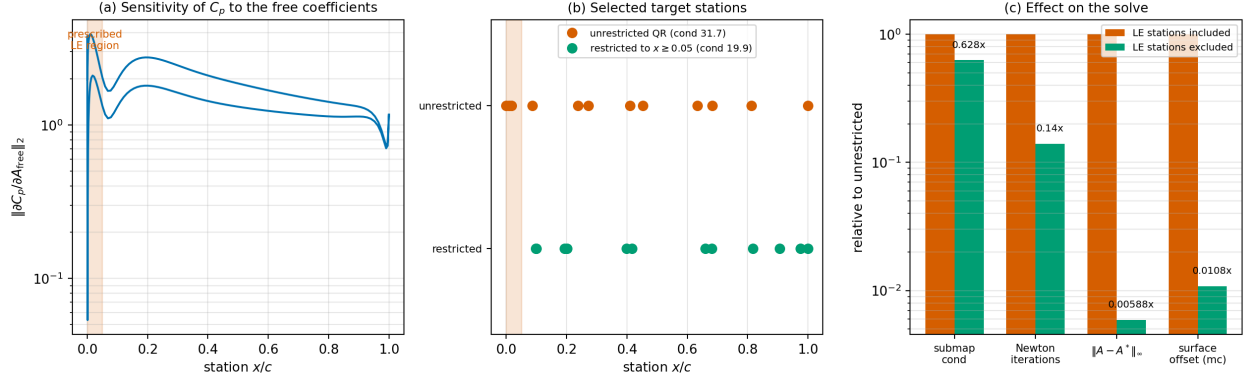


Figure 11: Why target stations must be kept out of a prescribed leading edge: (a) pressure sensitivity to the free coefficients across the chord, prescribed region shaded; (b) stations chosen by unrestricted against restricted QR; (c) the effect on the solve, relative to the unrestricted case.

the condition number of the assembled extended Jacobian, and the condition number of the Bernstein Gram matrix used for fitting.

Table 7: Bernstein order sweep, $n = 4$ to 12 per side. Iteration count is flat through $n = 10$ and then jumps at $n = 12$, while all three conditioning measures climb monotonically.

n	Iterations	$\ A - A^*\ _\infty$	Submap cond	cond(J) at iteration 0	Gram cond
4	3	8.08×10^{-12}	10.6	6.46×10^7	498
6	4	1.59×10^{-10}	20.4	8.46×10^7	7,327
8	4	1.08×10^{-11}	90.3	1.57×10^8	108,538
10	4	1.46×10^{-10}	592	4.52×10^8	1,626,188
12	21	6.31×10^{-10}	1,144	2.07×10^9	24,579,174

Iteration count is flat at three to four through $n = 10$, then jumps to 21 at $n = 12$, missing the nine-iteration budget. This is a genuine cliff, not gradual degradation, though it tracks, without being identical to, all three independently-measured conditioning series, which climb monotonically with n throughout the sweep. The Gram condition number grows by a factor of approximately 15 for every increment of two in n , that is, close to 3.9^n , which is the 4^n growth anticipated a priori for a Bernstein basis and one of the cleanest quantitative confirmations in this study. Notably, $n = 12$ still reaches the accuracy criterion, at 6.31×10^{-10} against a 10^{-4} threshold; it misses only the iteration budget, which is itself informative, since it locates the degradation in convergence *speed* before final accuracy. Figure 12 plots the sweep, with the three conditioning series on separate axes so that the reader is not invited to read them as one quantity.

5.4 Remaining single-factor ablations

Table 8 reports the remaining single-factor ablations against the $n = 8$ baseline. Three findings are worth drawing out. Forcing transition costs essentially nothing once the closure of Section 2.6 is correct, which is what makes it usable as a default rather than a last resort. Freeing the angle of attack costs one extra iteration and two orders of magnitude of final accuracy, consistent with the camber–incidence null direction identified in Section 5.1 and the reason incidence is held fixed

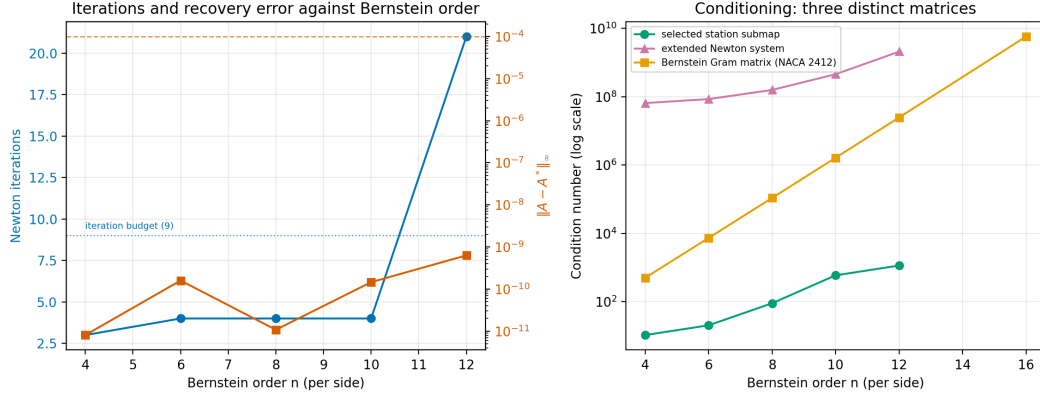


Figure 12: Iterations and recovery error against Bernstein order n , with the three distinct conditioning series plotted separately. The Gram series is carried to $n = 16$, two orders beyond the runs of Table 7.

for self-consistency tests. And skipping the pre-solve costs one extra iteration without costing correctness, which is the observation Section 5.7 develops.

Table 8: Single-factor ablations against the $n = 8$ baseline.

Factor varied	Iterations	$\ A - A^*\ _\infty$	Comment
Transition free rather than forced	4	1.28×10^{-11}	no material difference; the mitigation is close to free once applied
Incidence free rather than fixed	5	2.31×10^{-9}	one extra iteration, consistent with the camber–incidence null direction
Perturbed rather than pre-solved start	5	2.12×10^{-11}	43 against 120 counted solves (Section 5.7)
Leading edge free rather than prescribed	4	6.50×10^{-10}	submap condition roughly doubles, 195.6 against 90.3; still six orders inside the criterion
Deliberately over-determined by one row	0	not applicable	Eq. (9) violated; fails at construction with an explicit message
Deliberately under-determined by one row	0	not applicable	Eq. (9) violated; fails at construction with an explicit message

The two deliberately mis-squared configurations fail at construction time with the explicit, correct error strings quoted in Section 2.3 and zero Newton iterations spent. The degrees-of-freedom guard catches over- and under-determination cleanly, with no silent corruption and no wrong answer, which is the behaviour the guard exists to produce and the reason it is an assertion rather than a diagnostic.

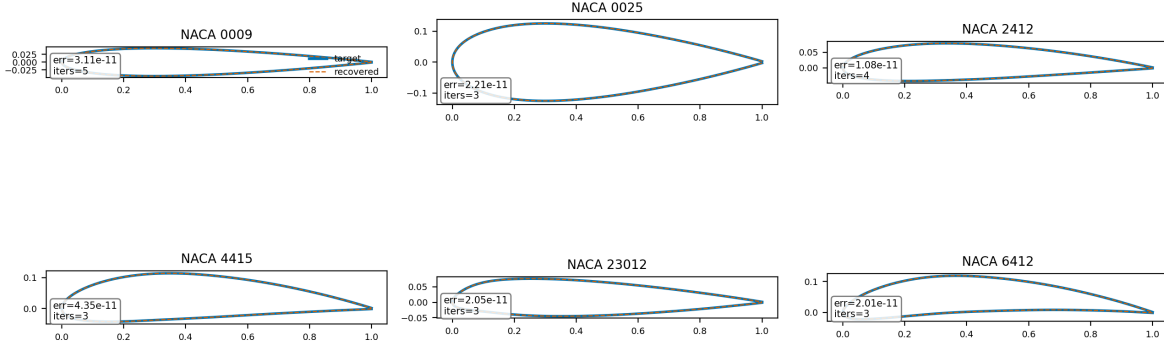
5.5 Generalisation: the NACA panel

Table 9 adds two additional single-airfoil runs, on NACA 0012 and 4415, to the 2412 baseline reported throughout Section 4. That three-airfoil comparison is one run per section and is explicitly not a generalisation claim; it is included because its submap conditioning and model gap vary with camber and thickness in a way the panel results then confirm at scale. It was superseded by the pre-registered panel evaluation itself, run after the ablation matrix closed: the configuration these ablations identify, applied to a 20-section NACA panel spanning 6 to 25 percent thickness and 0 to 6 percent camber across both four- and five-digit families.

Table 9: Three-airfoil detail: the baseline and two additional sections.

Airfoil	Iterations	$\ A - A^*\ _\infty$	Submap condition	Model gap
NACA 2412 (baseline)	4	1.08×10^{-11}	90.3	0.110
NACA 0012	4	3.15×10^{-11}	83.9	0.083
NACA 4415	3	4.35×10^{-11}	117.9	0.163

Result. All 18 generable sections are recovered, to $\|A - A^*\|_\infty \leq 1.51 \times 10^{-10}$ in three to seven Newton iterations each. Two of the 20 pre-registered sections returned no successful result and are excluded under the pre-registered exclusion rules rather than counted as failures. For the 6 percent-thick symmetric section, the forced-trip *target* generation itself fails to converge at a 5 percent trip location, so no inverse problem was ever posed; that is a direct-solve non-convergence exclusion, exactly the pre-registered rule’s intended case, and not an architecture failure. For one five-digit section, the vendor geometry generator does not implement the required mean-line family, which is a defect in the pre-registered target list rather than a solver failure. The 18 recovered sections span 8 to 25 percent thickness. Figure 13 shows six recovered sections spanning most of that range: thin symmetric, thick symmetric, the cambered baseline, a thick cambered section, a standard five-digit-camber section, and a high-camber section. Target and recovered outlines are visually indistinguishable at this scale, consistent with the coefficient-level agreement quoted above.

**Figure 13:** Six representative recovered NACA panel sections, target against recovered geometry, annotated with each section’s recovery error and iteration count.

Why the pre-registered criterion still fails at this panel size. At 18 successes out of 18 eligible trials, the Wilson 95 percent lower bound is 0.824, below the pre-registered criterion of 0.9, even though every eligible section succeeded. This is not a solver shortfall. A Wilson lower bound of 0.9 is mathematically unattainable at 18 or 20 trials regardless of outcome: the smallest all-success sample whose Wilson lower bound clears 0.9 is 35, at 0.901, and 29 trials reach only 0.883. The pre-registered target and the pre-registered panel size are therefore mutually inconsistent, which is a pre-registration defect. Treating the excluded thin section as a failure rather than an exclusion instead gives 18 of 19, a fraction of 0.947 with Wilson lower bound 0.754, still short of 0.9 for the same sample-size reason. On the NACA panel alone, the architecture converges and recovers the true coefficients on every section the problem could legitimately be posed on, and the formal statistical criterion fails on interval arithmetic rather than on solver performance. Section 5.6 reports the pre-registered extension, sized to resolve that limit, and gives the combined verdict across both

panels.

5.6 Generalisation: the UIUC panel

Tables 10 and 11 report the pre-registered second panel. Sections were drawn from a 123-file corpus of the UIUC airfoil coordinate database, stratified by thickness and camber decile, after a pre-filter removed six sections whose $n = 8$ CST fit exceeds the representation tolerance, leaving 117 configurations.

Table 10: UIUC panel outcome by population.

Population	Definition	Count
Configurations	Sections with a generated configuration	117
No inverse problem posed	No usable starting point existed: for 23 sections the target solve did not converge, and for one the flow solve at the initial geometry failed	24
Attempted	Configurations where a target existed and the inverse was run	93
Converged	Newton reached the residual tolerance	83
Attempt failures	Solver or geometry faults during the attempt	10

Result. Accuracy holds on every converged section; the composite criterion fails on iteration count. Of the 117 configurations, 24 pose no inverse problem, 83 of the remaining 93 converge, and 10 fail during the attempt. All 83 converged sections recover the free coefficients to better than 10^{-4} , a success fraction of 1.000 with Wilson 95 percent lower bound 0.956. The iteration-budget criterion of nine Newton iterations holds for 73 of the 83, a fraction of 0.880 with lower bound 0.792. The pre-registered composite criterion, accuracy and iteration budget together, is therefore 0.880 among converged sections and 0.785 among attempted sections, with lower bounds 0.792 and 0.691, both below the pre-registered threshold of 0.9.

Table 11: UIUC panel criteria and Wilson 95 percent lower bounds.

Criterion	Result	Wilson 95% lower bound
Coefficient accuracy, among converged	83/83 = 1.000	0.956
Iteration budget of nine, among converged	73/83 = 0.880	0.792
Both criteria, among converged	73/83 = 0.880	0.792
Both criteria, among attempted	73/93 = 0.785	0.691

Ten converged sections exceed the iteration budget, requiring between 10 and 50 Newton iterations while still recovering coefficients to between 1×10^{-11} and 3×10^{-10} , the same accuracy band as the sections that converge within budget. The single-digit iteration expectation was calibrated on NACA sections (Section 5.5). It does not transfer to the wider UIUC geometry space, where sections with unusual camber or thickness distributions converge more slowly without loss of final accuracy. Figure 14 shows both facts at once: the right panel is the iteration distribution, strongly peaked at three to five with a thin tail running to the budget ceiling, and the left panel places every section of the panel in the thickness–camber plane, coloured by outcome.

Exclusions and the bias they carry. Table 12 lists the exclusion classes spanning the 24 sections for which no inverse problem was posed and the 10 attempt failures. The two target-generation

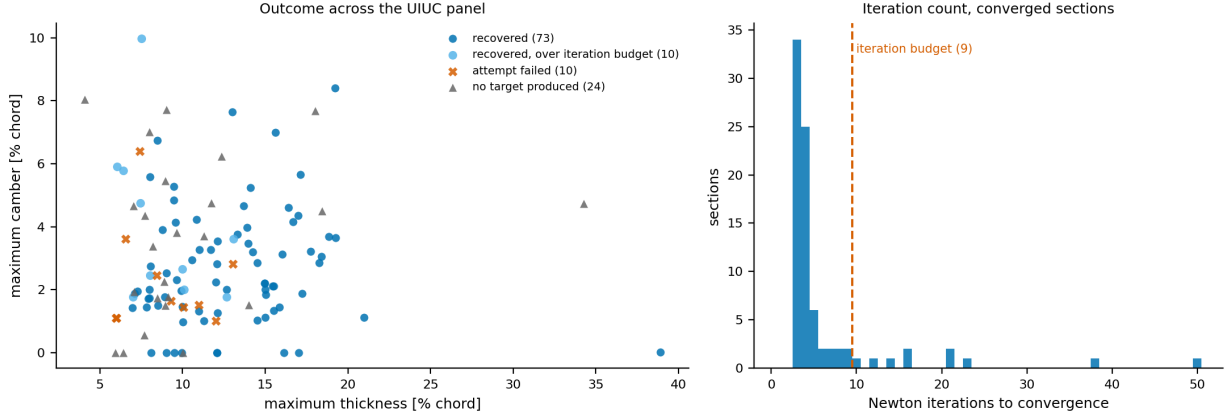


Figure 14: The UIUC panel by outcome: every section in the thickness–camber plane (left) and the iteration distribution among converged sections (right).

classes account for 23 of the 24 not-posed sections: the flow solver cannot produce a converged solution for the section itself, so no target exists and the inverse formulation is never exercised. Three sections fail the vendor’s wake-direction assertion after the minimum-trailing-edge-gap correction that resolved the sharp-trailing-edge case; their trailing-edge gap is already above the minimum, so the cause is distinct from the case that correction addressed, and this is recorded as an open item in Section 7.3 rather than resolved here.

Table 12: UIUC panel exclusion classes.

Exclusion class	Count
Target solve did not converge, natural transition	16
Target solve did not converge, forced transition	7
Solver stagnation during the inverse attempt	3
Trailing-edge geometry rejected by the flow solver	3
Other faults during the attempt	3
Runtime error during the attempt	1
Flow solve at the initial geometry did not converge	1

Those 34 excluded sections are 29 percent of the panel, and the honest reading of the 83-of-83 accuracy figure depends on whether they are geometrically representative. They are not. The excluded sections have a median maximum thickness of 9.0 percent chord against 12.1 percent for the converged population, and 65 percent of them are thinner than 10 percent chord against 36 percent of the converged sections; the not-posed group is also more cambered on average. This is visible directly in Figure 14, where the excluded markers cluster toward the thin and thin-and-cambered corner of the plane. The correct statement is therefore not that the architecture recovers any UIUC section, but that on sections for which this flow solver can produce a converged target at all, it recovers the coefficients to well inside the accuracy criterion, and that the sections it cannot produce a target for are systematically the thin ones. The exclusion is a property of the host analysis solver rather than of the inverse formulation, since the failures occur before any inverse problem is posed, but it bounds what the panel result can be claimed to demonstrate.

Combined verdict across both panels. Accuracy holds on every section that converged: 18 of 18 eligible NACA sections and 83 of 83 converged UIUC sections recover the free coefficients to better than 10^{-4} , with observed errors clustering between 10^{-11} and 10^{-9} . The pre-registered composite criterion, which also required completion within nine Newton iterations, is not met on the UIUC panel: its Wilson lower bound is 0.792 among converged sections and 0.691 among attempted sections, both below the pre-registered 0.9. The cause is the iteration budget, not accuracy, and the budget was calibrated on a NACA family narrower than the population it was then applied to. The recovery hypothesis is therefore supported on accuracy across both panels and unsupported on the composite criterion, with the failure isolated to the iteration-count component.

5.7 Computational cost against nested optimisation

The corrected headline. A reduction of at least hundredfold in flow solves was pre-registered; it is *rejected* under both fair-paired controls actually run. Table 13 pairs each monolithic run against a nested Levenberg–Marquardt control sharing the same initialisation strategy, so that the comparison is like-for-like, and reports three currencies rather than one.

Table 13: Cost comparison, monolithic against nested Levenberg–Marquardt control, in three currencies. Both arms of a pairing share an initialisation strategy.

Pairing	Monolithic solves	Nested solves	Solve ratio	Wall-clock ratio
Pre-solve initialisation	120	373	$3.1\times$	$3.4\times$
Perturbed (cold) start	43	347	$8.1\times$	$3.5\times$

Why three currencies and not one. The counted quantity, *flow-solve equivalents*, weights a converged viscous solve and a single inviscid linear solve equally, and it excludes the residual evaluations that the finite-difference Jacobian of Section 2.4 spends inside each Newton step. Both conventions favour the monolithic side. The nested control’s every residual call is a full converged viscous solve, counted once; the monolithic method’s residual evaluations, 100 and 133 in the two paired runs, are not counted at all, though each rebuilds the geometry and re-forms the panel influence system. Wall-clock time is immune to both conventions, and Figure 15 shows all three side by side. The measured wall-clock advantage is $3.4\times$ and $3.5\times$ on the two pairings, essentially identical, against solve-count ratios of $3.1\times$ and $8.1\times$. The $8.1\times$ figure is therefore a currency artifact: dropping the pre-solve removes 77 comparatively cheap inviscid solves from the monolithic count without removing the corresponding work. This paper reports the solve-count range because it was the pre-registered currency, and treats the wall-clock figure of approximately $3.4\times$ as the conservative statement of the result.

The nested control is a Levenberg–Marquardt solve from `scipy.optimize.least_squares` with function, step and gradient tolerances all at 10^{-8} and a per-variable scaling taken from the initial guess, floored at 10^{-3} [Virtanen et al., 2020], over the same 16 free coefficients. It converged in both pairings on the step tolerance, and it is a competent modern solver rather than a weak strawman: the cold-start case takes six outer iterations, each costing multiple flow solves for its own finite-difference Jacobian columns. It also reaches essentially the same final accuracy as the monolithic solve, 2.29×10^{-11} and 2.13×10^{-11} on the two pairings against 1.08×10^{-11} and 2.12×10^{-11} ; the difference between the two architectures is cost and determinism, not achievable precision on this class of target. One asymmetry remains in the pairing: the control assigns its target pressure by nearest-node remapping, with a residual correspondence error of 2.0×10^{-4} in chord fraction, while

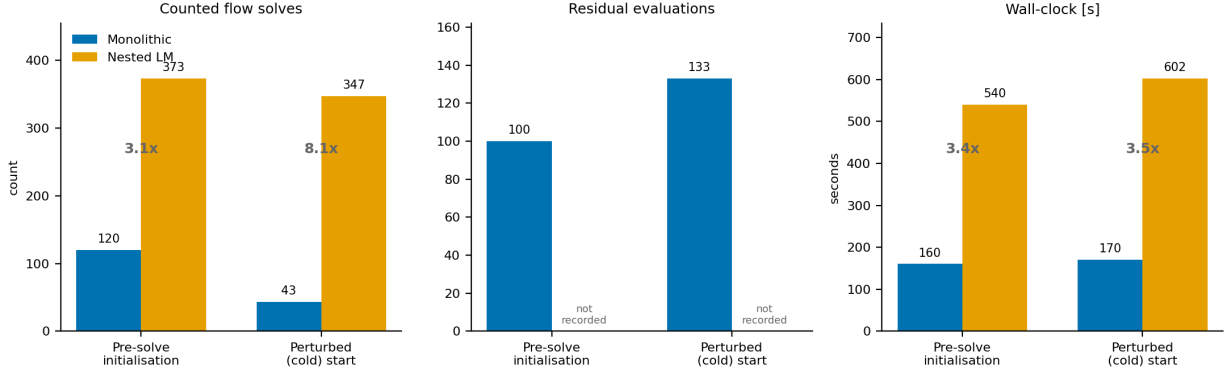


Figure 15: Cost of the monolithic solve against its fair-paired nested control in three currencies: counted flow solves, residual evaluations, and measured wall-clock time.

the monolithic arm of the same era used the same convention. Both arms therefore share the older addressing, but a reader comparing against Section 4’s interpolated-addressing run should not mix the two.

The summary sentence the evidence supports is that the monolithic architecture requires 3.1 to 8.1× fewer counted flow solves, and 3.4 to 3.5× less wall-clock time, than a fairly-paired, well-tuned nested Levenberg–Marquardt baseline, substantially less than the two-to-three-orders-of-magnitude reduction hypothesised a priori, while the qualitative advantages of determinism, an exact analytic Jacobian for the constraint rows, and per-iteration failure-mode diagnostics persist independent of this ratio.

Pre-solve or station selection as the uniqueness guard. Skipping the pre-solve, with QR station selection retained, gives 43 flow-solve equivalents in five iterations at an error of 2.12×10^{-11} , with a submap condition of 89.8: essentially identical to the baseline’s 90.3, since station selection is a property of the stations, computed once from the sensitivity matrix, and not of the initial guess. Retaining the pre-solve gives 120 flow-solve equivalents in four iterations at 1.08×10^{-11} and a submap condition of 90.3. The pre-solve consumes roughly 77 flow-solve equivalents, about 64 percent of the baseline total, and buys one Newton iteration and a better starting residual. In wall-clock terms the two are indistinguishable, at 160 against 170 seconds, so the honest conclusion is that the pre-solve is close to cost-neutral rather than expensive, and that it is not what supplies the uniqueness guarantee. That comes from QR station selection alone. This has *not* been verified in the sharper crossed form that would settle it, a perturbed start combined with evenly-spaced station selection, and it is flagged as untested rather than as tested.

5.8 Convergence behaviour

The three-point local order estimator of Eq. (17) is applied only where at least three residual values exist strictly above the finite-difference floor near 10^{-9} identified in Section 2.4; below that floor the sequence measures the Jacobian’s own noise rather than the iteration’s convergence. Appendix D states the estimator, the floor rule and the residual block each estimate is computed on. Table 14 reports the estimates for the five runs with enough above-floor residuals to support one.

The median estimate is 2.15 over these five runs, meeting the pre-registered target of 1.8, but the spread from 1.09 to 4.24 is wide enough that the median should not be read as a measurement of a single underlying rate. The remaining seven converged runs cannot be scored by this method because

Table 14: Local convergence order over the above-floor tail, for the five runs where three clean residuals exist.

Run	Estimated order	Note
Incidence free	1.09	the slowest of the set; consistent with the camber-incidence null direction
Perturbed start	2.08	
$n = 6$	2.46	
Transition free	2.15	
$n = 12$	4.24	late-tail estimate after a long stalled phase; read as clearly superlinear, not as a literal quartic order

they reach the residual floor in only two post-initial steps, too fast for a three-point estimator to have three clean points. That is itself consistent with, though it does not formally confirm, fast superlinear convergence: the baseline, for example, drops from 5.9×10^{-4} to 3.1×10^{-6} and then to 1.0×10^{-10} , contracting by factors of about 190 and then 30,000, a pattern no linear rate produces.

The reference run behaves differently, and the reason is instructive. The self-consistency run of Section 4, which uses interpolated station addressing, contracts by about 1/30 per step on average, between 1/13 and 1/130, without accelerating: its above-floor local order estimate is 0.53, and the three-point estimates across its tail range from 0.5 to 1.6. It is fast, dropping seven decades in five steps, but it is not showing a quadratic tail. Figure 16 makes the distinction visible: the left panel is the residual history and the right panel is the per-step contraction factor, which falls over the tail for the faster node-indexed runs and does not for the interpolated one. The explanation follows directly from Section 2.3. Under node-index addressing $\partial T / \partial A$ is structurally zero and the assembled Jacobian is exact up to the finite-difference error in $\partial R / \partial A$; under interpolated addressing it is small but nonzero and is neglected, which makes the iteration an inexact Newton method with a bounded Jacobian error and therefore a linear asymptotic rate with a small contraction constant. The practical consequence is minor, since six iterations suffice either way, but the claim that should be made is fast superlinear convergence in practice, limited by the finite-difference Jacobian and the neglected interpolation term, and not guaranteed quadratic convergence. Guaranteed quadratic convergence would require an exact, Lipschitz, nonsingular Jacobian, and this implementation has none of the three by construction.

5.9 Comparative positioning against other inverse-design approaches

Tables 15 and 16 place this paper’s architecture against five other approaches to the same underlying problem, recovering a geometry that produces a target surface pressure or speed distribution, each representing a structurally distinct formulation class. The comparison is split across two tables because its two halves are not commensurable in the same way. Table 15 collects the quantitative axes, where the cost column is reported in whatever unit each formulation class naturally exposes, and Table 16 collects the structural guarantees, where the classes separate sharply and no numbers are involved at all. Rows (i) and (ii) are this paper’s own measured, fair-paired numbers (Section 5.7); rows (iii) to (vi) are drawn from the cited literature and are marked qualitative wherever the source does not report a directly comparable metric. No number in either table is asserted unless it is a

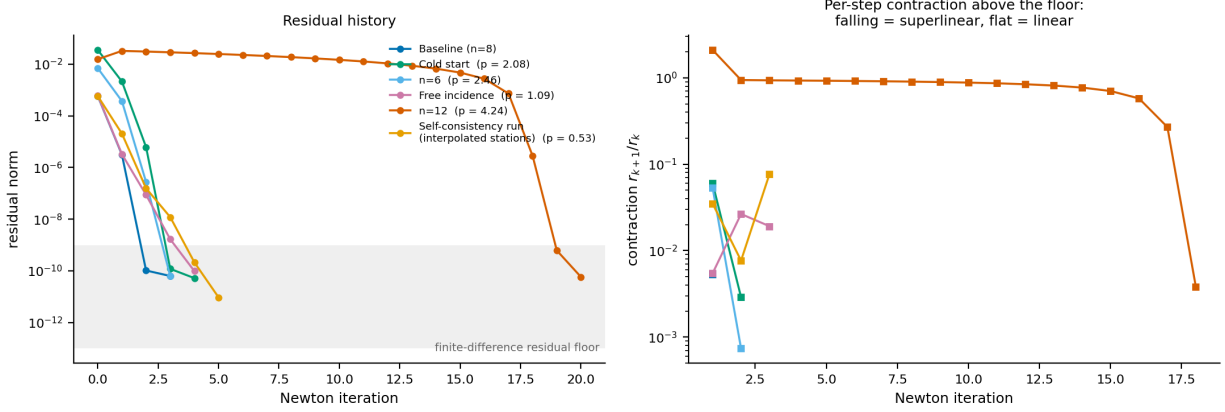


Figure 16: Residual histories with the local order estimated over the above-floor tail (left), and the per-step contraction factor (right), which falls for node-indexed runs and stays flat for the interpolated one.

repository artifact from rows (i) or (ii), or attributed to a specific cited source.

Three patterns follow. First, only rows (i) and (iii) are architecturally root-finding rather than optimisation or amortised inference. This distinction, argued in Section 1.1, is not cosmetic: a converged root-find has a residual-norm certificate that a converged optimisation or a sampled generative output does not. Second, the cost column spans roughly four orders of magnitude across the table, not because the methods disagree about a shared quantity, but because a flow solve means a different thing in each formulation class: a Newton linearisation, an outer least-squares iteration, an offline training epoch, or a primal and adjoint pair. This is exactly the currency-matching discipline Section 5.7 enforces for the one comparison, row (i) against row (ii), that this paper actually measured, and it is why rows (iii) to (vi) are reported qualitatively rather than omitted. Third, the availability column is not incidental context: it is the practical reason the MISES modal inverse mode of row (iii), despite being architecturally the closest relative of this paper’s contribution, has not seen the open comparative study this table attempts. A solver behind a licence gate cannot be the basis of independently reproducible comparison. That is also why `mfoil` [Fidkowski, 2022] rather than MISES is this paper’s testbed, and why the extension discussed in Section 8 targets porting the architecture *into* MISES rather than requiring readers to already hold a MISES licence to reproduce these results.

6 Released software artifacts

A method described only in prose is hard to check and harder to reuse. Three artifacts accompany this manuscript, and this section states what each contains, what it can and cannot do, and where the boundary between the deterministic engine and everything built on top of it lies. The boundary is the design constraint that makes the rest possible: the engine has no knowledge of the web layer, and every array crossing into it is the same array the experiments in Section 5 were run on. Figure 17 shows the arrangement.

6.1 The interactive application

A deployed web application exposes the method to a reader who does not want to install anything. It is a Python service wrapping the engine, with a browser front end, and it offers four views.

Table 15: Comparative positioning, quantitative axes. Cost is reported in the unit each formulation class naturally exposes.

Method	Formulation class	Cost per design	Demonstrated precision
(i) This work, monolithic CST–Newton	constrained root-finding, single square Newton system	43–120 counted solves, 160–170 s wall-clock, on the ablation-matrix runs of Section 5.7	$\ A - A^*\ _\infty$ at the 10^{-11} level on self-consistency (Table 4)
(ii) Nested Levenberg–Marquardt baseline [Virtanen et al., 2020]	outer-loop nonlinear least-squares optimisation	347–373 counted solves, 540–602 s wall-clock, on the same paired cases	$2.1\text{--}2.3 \times 10^{-11}$ against the same coefficient-level target, paired with row (i)
(iii) MISES modal inverse [Drela, 1990]	constrained root-finding, mode-shape amplitudes appended to the same global Newton system	one Newton solve per design, architecturally identical to row (i); exact count not published	qualitative: production-validated on transonic cascades over decades, no single benchmark number available
(iv) CST-smoothing inverse [Lane and Marshall, 2010]	iterative outer loop; CST used as a post-hoc smoother on a pressure-residual-driven shape correction, solver-agnostic	qualitative: multiple outer iterations, count depends on the wrapped solver and is not reported as a fixed number	qualitative: subsonic and transonic cases shown converging; no coefficient-recovery benchmark reported
(v) Learned generative inverse [Yilmaz and German, 2020, Deng et al., 2025, Wong et al., 2024]	amortised statistical inference, offline-trained generative or surrogate map	approximately zero online per design after training; training-set generation requires order 10^3 to 10^5 offline flow solves	a 34.4 percent precision improvement over Wasserstein-GAN baselines [Deng et al., 2025], relative rather than absolute; at least tenfold online inference speedup against direct analysis [Wong et al., 2024]
(vi) Adjoint target-pressure optimisation [Economou et al., 2016]	gradient-based PDE-constrained optimisation, primal and adjoint pairs per design cycle	qualitative: order 10 to 100 primal and adjoint pairs typical at this scale, case- and tolerance-dependent	qualitative: mature and widely validated on shape-optimisation benchmarks; not run in this study

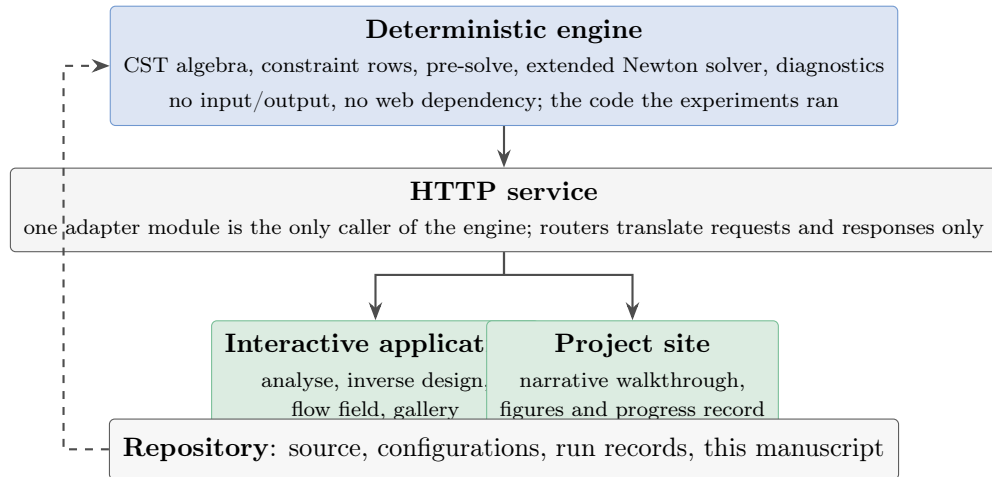


Figure 17: The released artifacts. The engine is the code the experiments ran; everything above it is an adapter, and nothing above it can change a result.

Table 16: Comparative positioning, structural axes. Determinism is the column on which the formulation classes separate most sharply.

Method	Determinism and uniqueness guard	Constraint handling	Licence and availability
(i) This work, monolithic CST–Newton	deterministic Newton; QR-pivoted station selection is an explicit, measured identifiability guard (Section 5.1)	exact linear equality rows $GA = b$ with a constant Jacobian block; implemented and tested, but not exercised by the measured results (Section 2.2)	open [Sathish, 2026]; the flow solver is MIT-licensed [Fidkowski, 2022]
(ii) Nested Levenberg–Marquardt baseline	deterministic given fixed initialisation and tolerances; no analogous uniqueness guard, relying on local convergence only	none built in; bounds and scaling only, with no analytic constraint rows in this configuration	open [Virtanen et al., 2020]
(iii) MISES modal inverse	deterministic Newton; no published uniqueness or identifiability study of mode count against target sampling analogous to Section 5.1	mode amplitudes are heuristic bump functions rather than analytic geometric functionals (Section 1.1)	distributed under a restricted academic and commercial licence; no open-source port exists
(iv) CST-smoothing inverse	deterministic given fixed settings; no formal uniqueness guard reported	no analytic constraint rows; the correction is smoothed but not constrained to a design-parameter manifold	method description only, solver-agnostic, with no released code cited
(v) Learned generative inverse	<i>not</i> deterministic for adversarial and diffusion variants, which sample stochastically; a trained surrogate is deterministic but approximate. No convergence guarantee or exact-recovery certificate exists for any variant	none intrinsic to generation; plausibility is learned from the training distribution rather than enforced algebraically	open-source code released with most cited papers, but a training database is required
(vi) Adjoint target-pressure optimisation	deterministic gradient descent to a <i>local</i> optimum; no uniqueness guard, which is the sense in which adjoint target-pressure matching remains optimisation rather than root-finding	general nonlinear constraints via adjoint sensitivities: flexible, but iterative rather than closed-form	open source under the LGPL [Economou et al., 2016]

Analyse runs a direct coupled solve for a section chosen from the same 123-file coordinate corpus the panel of Section 5.6 was drawn from, or from a NACA code, and reports the pressure distribution, the boundary-layer state, and the CST-derived engineering parameters, being leading-edge radius, trailing-edge wedge half-angles, maximum thickness and camber with their locations, and inscribed area, all evaluated in closed form from the coefficients by the same Beta-function machinery as Eq. (7) rather than by quadrature. *Inverse design* runs the monolithic solve itself, iteration by iteration, in either of two modes: a self-consistency solve against a target generated from a section’s own fit, which is the experiment of Section 4, or a solve against a target the user supplies as a pressure or edge-velocity curve. *Flow field* renders the inviscid velocity and pressure field around the section. *Gallery* replays the archived results and figures this manuscript reports.

Two behaviours of the user-supplied-target mode are worth recording here, because they are the product-side evidence for Section 5.2’s finding. First, the realisability verdict is always reported and never blocks: a target outside the CST-representable manifold produces a warning carrying the inviscid-consistent metric of Section 2.7, not an error, which is the reporting rule of that section made into interface behaviour. Second, the same station-restriction rule the panel runs use is applied here, and it was in fact found here first. On a user-defined viscous target the unrestricted candidate set converged in 14 iterations to a surface 0.86 millichords from the target, while restricting candidates to stations aft of the prescribed nose converged in 8 to 0.017 millichords, a fiftyfold improvement in recovered shape at an almost unchanged residual. Recovery on such a target is reported and asserted on the *shape* rather than on the coefficients, because the CST basis is ill-conditioned in A (Section 5.3) and coefficient vectors differing by 1.8×10^{-2} describe surfaces differing by less than half a millichord; against a target that only ever constrained pressure, the coefficient norm is not a meaningful accuracy statement.

Two limitations of the deployment are stated so that a reader who tries it is not misled. The service runs one inverse solve at a time by construction, because the forced-transition closure of Section 2.6 replaces module-level functions in the vendored solver and is therefore process-global rather than instance-scoped; a second submission waits rather than running concurrently. And the rendered flow field is inviscid, since the displacement-thickness source contribution is not yet added to it, while the boundary-layer quantities displayed beside it come from the fully coupled solve.

6.2 The project site and the repository

A project site presents the method as a narrative walkthrough, from the problem statement through the linearity argument, the square system, the falsifiable test, the identifiability finding and the conditioning cliff, using the same figures as this manuscript. It exists because the argument of Section 2.1 is visual before it is algebraic, and a reader deciding whether to spend time on the formulation is better served by the figures than by the equations.

The repository carries the engine, the configuration files that define every run reported here, the archived run records those runs produced, the figure scripts, the test suite, and the source of this manuscript. Appendix E states the reproduction procedure and the software environment; Appendix F maps each figure and table in this manuscript to the record it was produced from. The vendored flow solver is included unmodified under its own licence and is never edited; all access to it passes through a single adapter module, which is what made the forced-transition closure of Section 2.6 implementable without a fork.

7 Discussion

7.1 A common structure across the four mitigations

The mitigations adopted in this paper are not independent engineering fixes. Each was arrived at separately, from direct numerical evidence, a stalled residual, a spiked row norm, a badly conditioned submap, and the common structure was recognised only afterwards. Recording it is worth a paragraph, because it suggests where to look first when the architecture is carried into a solver where the specific failures differ.

Three of the four resolve a tension by separating the two things in conflict rather than by compromising between them. The transition closure of Section 2.6 separates them in *time*: the conflict is between a smooth, Newton-tractable residual and an accurate transition model, and it is resolved by phase, forcing transition during the inverse iterations and releasing it afterwards, with a verification solve confirming that nothing was lost. Nothing is traded away in the end state; the kink is removed exactly when Newton needs it removed. The prescribed leading edge separates them in *space*: the conflict is between design freedom, invert everywhere, and row conditioning, since the residual rows near the nose blow up, and it is resolved by prescribing the first few percent of chord geometrically and inverting only aft of it. That does cost design freedom, and Section 7.3 records the cost rather than explaining it away. The station-restriction rule of Section 5.2 separates them along the same axis, one level down: having removed a region from the design space, the sampling must be removed from it too.

The remaining two resolve their tension by improving the quality of what is already there rather than by adding machinery. For the basis-conditioning mode, the obvious fix, Tikhonov regularisation, turns a determined solve back into a least-squares problem, which is the optimisation this architecture exists to avoid; the discipline adopted instead is to keep the basis order inside its well-conditioned regime, or to orthogonalise the basis, rather than regularise a badly-conditioned one. For station selection, adding more target equations would over-determine the square system, and accepting whatever stations are convenient is what produced the wrong root of Section 5.1; the resolution reuses an operator the pre-solve had already built to choose which M rows are maximally informative. Both are instances of the same instinct: when a determined formulation is the point, the fix must not add degrees of freedom.

7.2 Threats to validity

The most important limitation of this study is structural and is stated first because it conditions how every accuracy number in Section 5 should be read. No result reported here recovers a geometry from a target that was produced independently of the method. In every case, the self-consistency test, the ablation matrix and both panels, the target pressure distribution is generated by the same solver, at the same paneling, from a coefficient vector that is exactly representable in the same CST basis at the same order. Recovery errors at the 10^{-11} level are therefore measurements of *identifiability and conditioning*, showing that the sampled square system has the generating design as its unique reachable root and that the Newton iteration finds it, and not measurements of accuracy against independent data. This is deliberate: a self-generated target is the only construction under which a failure isolates cleanly to one of the Table 2 failure modes rather than to target infeasibility, which is what makes the test falsifiable in the first place. But it means the reported precision does not transfer to a target drawn from experiment, from a finer discretisation, or from a different solver. The nearest thing to evidence outside this regime is the user-supplied-target path described in Section 6, which recovers shape to a fraction of a millichord while leaving coefficients to differ at the 10^{-2} level. Even that solve was run against a self-generated target, and its coefficient gap

is attributable to the inviscid pre-solve meeting a viscous target and to the basis conditioning of Section 5.3 rather than to target independence, so it bounds nothing. A genuinely independent target has not been attempted, and doing so is the first item of future work.

Two further gaps in the evidence follow from the same source. The realisability guard of Section 2.7 is never exercised on an unrealisable target: measured realisability on the reference configuration is 3.90×10^{-4} against a threshold of 0.05, because a self-generated target is realisable by construction, so the guard’s discriminating power is asserted rather than demonstrated. And there is no robustness study: real inverse targets are noisy, over- or under-specified, and nothing here tests target perturbation or a target that is only nearly realisable. Given that the identifiability mechanism of Section 5.1 is a statement about how much information the sampled rows carry, a sweep of recovery error against target-noise amplitude is the single cheapest experiment that would sharpen this paper’s central finding, and it has not been run.

The discretisation is also fixed throughout. Every result uses the same panel count, and for a method whose target rows attach to surface stations and whose flow-block Jacobian is a finite difference over geometry, sensitivity to panel count is a first-order question that this study does not answer.

7.3 Scope limits

The ablation evidence is single-airfoil and single-run. The order sweep, the station-selection comparison, the remaining single-factor ablations of Section 5.4 and the cost comparison of Section 5.7 are all on NACA 2412 at a fixed seed, one run per configuration, with no confidence interval reported for any of them. That caveat does not extend to the recovery claim itself, which was upgraded to panel-level evidence across both the 20-section NACA panel and the 117-section UIUC panel (Sections 5.5 and 5.6); but the ablation-level findings remain single-airfoil and are not extended to either panel here.

The panel evidence carries a selection effect, quantified in Section 5.6. Twenty-nine percent of the UIUC panel is excluded, overwhelmingly because the host analysis solver cannot converge a target for the section at all, and the excluded sections are systematically thinner than the converged ones. The exclusion arises before any inverse problem is posed and is therefore a property of the host solver rather than of the formulation, but the accuracy claim is conditioned on it. Three further sections fail the host solver’s wake-direction assertion for a cause unrelated to the sharp-trailing-edge case that an earlier correction addressed; their trailing-edge gap is already above the minimum, and this remains an open item.

The flow regime is narrow. All results are incompressible: the configuration sets free-stream Mach number to zero, so the solver’s Kármán–Tsien compressibility correction, which would in any case be valid only below the critical Mach number, is never exercised. Compressible and transonic behaviour is untested, as is anything cascade-specific, since there is no periodicity, no pitch or stagger, and no streamtube contraction. All runs share a single operating condition, one Reynolds number and one incidence, so no claim here generalises across flow regimes. The hypothesis under test is a claim about solver *architecture* rather than about turbine aerodynamics, and is answered as well on an isolated airfoil as on a cascade; but the regimes the architecture was ultimately built for are genuinely untested.

Two limits internal to the formulation were flagged in Section 1.2 and are repeated here because they bound the contribution rather than the evidence. The constraint rows, which are half of the novelty claim, are implemented and unit-tested but not exercised by any measured result, so wherever an exact analytic constraint-row Jacobian is offered as an advantage over the nested baseline, it is an advantage of the formulation and not one this study has measured, in a comparison where neither side

carried an active constraint. And the flow-block Jacobian is obtained by finite differences rather than from the exact constant geometric sensitivity, because the host solver exposes no complete geometric sensitivity of its own residual; the exact sensitivity is used where it can be, in the constraint rows and the pre-solve, and Section 5.8 measures what the finite-difference substitute costs in convergence rate.

Finally, the novelty claim itself rests on a negative result established by open literature search rather than by an exhaustive search of the aerospace society digital collections. No published work is known that couples a Newton-coupled inverse solver directly to CST coefficients as Newton unknowns, as opposed to CST as a post-hoc smoother, but that is a statement about what was found rather than about what exists.

7.4 Why the pre-registered cost threshold was missed

The pre-registered hundredfold claim is not simply missed by the measured 3.1 to $8.1\times$ in counted solves, or by the $3.4\times$ in wall-clock time. The two fair-paired controls of Section 5.7 give a plausible account of *why* it lands where it does. A Levenberg–Marquardt solve with all three tolerances at 10^{-8} and per-variable scaling taken from the initial guess, over the same 16 well-conditioned free coefficients, is a genuinely strong comparator. It converges the cold-start case in six outer iterations, which is nothing like the surrogate-in-the-loop nested-optimisation baselines that a hundredfold figure typically describes in the older aerodynamic shape-optimisation literature the original framing drew on. The pre-registered threshold plausibly encodes an outdated class of nested baseline. Measuring against a strong modern gradient-based one and reporting the resulting ratio directly, in three currencies rather than the most favourable one, is a more informative comparison than either confirming or missing the original number.

8 Conclusions

Appending CST coefficients as unknowns to a coupled viscous/inviscid Newton solver, with geometric design constraints expressed as the linear rows CST’s own linearity makes available, converts constrained inverse airfoil design from a search over an objective function into a single square, determined root-find. The falsifiable self-consistency test of Section 4 passes with a wide margin on every accuracy and iteration-count criterion, including a release-and-verify check the original protocol had not required, though its convergence rate is fast superlinear rather than quadratic (Section 5.8). The ablation matrix of Section 5 confirms the architecture’s viability across three NACA sections and Bernstein orders up to a conditioning cliff at $n = 12$, and the two pre-registered panels extend the recovery claim to 18 of 18 eligible NACA sections and 83 of 83 converged UIUC sections, subject to the selection effect Section 5.6 quantifies.

Two of the study’s results correct the hypothesis stated in Section 1.3. The cost advantage over a competently-tuned nested baseline is 3.1 to $8.1\times$ in counted flow solves and 3.4 to $3.5\times$ in wall-clock time, not the two to three orders of magnitude assumed a priori. And the convergence behaviour is fast superlinear rather than guaranteed quadratic, limited by the finite-difference flow-block Jacobian and, under interpolated station addressing, by a neglected term in the target rows. Against those corrections stands a methodological contribution the original hypothesis did not anticipate: QR-pivoted station selection as the identifiability guard for square inverse systems, without which the same solver converges cleanly to a wrong root while reporting success. The paper’s central results are therefore a falsified numeric hypothesis and a strengthened, unanticipated qualitative one.

Future work. The single most valuable next experiment is the one Section 7.2 names: recovering a geometry from a target this method did not generate, whether from a finer discretisation, a perturbed distribution, or a published pressure distribution, together with a sweep of recovery error against target noise. Until that is run, the precision reported here is a statement about identifiability rather than about accuracy.

Beyond it, two extensions are pre-planned and neither is yet built. Cascade periodicity substitutes a periodic kernel for the isolated-airfoil panel-influence kernel, replacing $\ln(z - z_0)$ with $\ln[\sin(\pi(z - z_0)/s)]$ for pitch s , which reduces to the isolated-airfoil kernel as $s \rightarrow \infty$ and therefore comes with a free regression test; it would be validated against exact cascade solutions before any turbine-cascade inverse-design claim is made. Carrying the architecture into MISES itself uses that solver’s modal inverse mode. Because CST’s perturbation modes are identical to its own basis functions, a direct consequence of the linearity of Section 2.1, this requires writing $C(\psi)S_i(\psi)$ into the solver’s mode-file format with no modification to its source, and would demonstrate the hypothesis in a transonic, viscous, real-cascade solver at near-zero implementation cost if that route succeeds.

The gap between what this paper can currently claim and what was originally pre-registered is threefold. The single-run ablation evidence has not been extended to panel level. Three UIUC sections still fail the host solver’s wake-direction assertion for a cause unrelated to sharp trailing edges. And the constraint rows, which are half of the formulation’s novelty claim, remain implemented and tested but unexercised by any measured result.

Data and code availability

All source, configurations, run manifests and figure-generation scripts are available at <https://github.com/SharathSPhD/CINS>, together with the deployed application and the project site described in Section 6. Every figure in this manuscript is regenerable from the archived configuration and run record that Appendix F names for it, under the procedure of Appendix E. The flow solver is `mfoil`, used unmodified as a vendored dependency.

Acknowledgements

The author thanks Krzysztof Fidkowski for releasing `mfoil` under the MIT licence, which made a Newton-coupled viscous solver with an accessible global system available for this work, and Michael Selig and the University of Illinois at Urbana–Champaign for maintaining the public airfoil coordinate database from which the evaluation panel of Section 5.6 was drawn.

Appendix A. Beta-function area-row derivation

Section 2.2 states the inscribed-area constraint row as a closed-form Beta function without deriving it; this appendix supplies the algebra and the two-surface assembly. The per-surface area under Eq. (4) is

$$\mathcal{A}_{\text{surf}} = \int_0^1 \zeta(\psi) d\psi = \int_0^1 \left[C(\psi) \sum_i A_i S_i(\psi) + \psi \zeta_T \right] d\psi = \sum_i A_i \int_0^1 C(\psi) S_i(\psi) d\psi + \frac{\zeta_T}{2}, \quad (11)$$

using linearity of the integral, itself a consequence of Eq. (4)’s linearity in A , and $\int_0^1 \psi d\psi = 1/2$ for the trailing-edge term. The remaining integral, per basis function $S_i(\psi) = K_i \psi^i (1 - \psi)^{n-i}$ with

$K_i = \binom{n}{i}$ and $C(\psi) = \psi^{N_1}(1 - \psi)^{N_2}$, is

$$\int_0^1 C(\psi) S_i(\psi) d\psi = K_i \int_0^1 \psi^{N_1} (1 - \psi)^{N_2} \psi^i (1 - \psi)^{n-i} d\psi \quad (12)$$

$$= K_i \int_0^1 \psi^{i+N_1} (1 - \psi)^{n-i+N_2} d\psi. \quad (13)$$

The Euler integral of the first kind is, by definition, $B(p, q) = \int_0^1 t^{p-1} (1 - t)^{q-1} dt$ for $\Re(p), \Re(q) > 0$. Matching exponents, $p - 1 = i + N_1$ so $p = i + N_1 + 1$, and $q - 1 = n - i + N_2$ so $q = n - i + N_2 + 1$, giving exactly Eq. (7):

$$\int_0^1 C(\psi) S_i(\psi) d\psi = K_i B(i + N_1 + 1, n - i + N_2 + 1). \quad (14)$$

For the default round-nose, sharp-tail class ($N_1 = 0.5, N_2 = 1.0$) this is $K_i B(i + 1.5, n - i + 2)$, always well-defined since $i + 1.5 > 0$ and $n - i + 2 > 0$ for every $i \in \{0, \dots, n\}$. No term in the sum is singular, despite the class function's own $\sqrt{\psi}$ non-analyticity at $\psi = 0$, which is Section 3's point that the leading-edge singularity affects flow-residual *rows* and not CST's own geometric *functionals*.

The enclosed area is the difference of the two surfaces, $\int_0^1 (\zeta_u - \zeta_l) d\psi$. Collecting one entry per coefficient therefore gives a *single* row g whose upper block carries the Beta-function values above, whose lower block carries the same values with the opposite sign, and whose trailing-edge contribution is $\frac{1}{2}(\zeta_{T,u} - \zeta_{T,l})$. The sign on the lower block is not cosmetic. Under the stored-sign convention of Section 2.1, lower-surface coefficients are already negative, and applying the upper-surface form to them produces a row that is silently wrong in exactly the way the trailing-edge row was once wrong.

The right-hand side b is the target inscribed area, or, for the self-consistency protocol of Section 4, the area of the CST fit itself, $b = g \cdot A^*$. This target-consistent convention is binding: the constraint right-hand side must always be built from what the reference geometry *actually has*, never from an idealised or rounded value, or the constraint row itself introduces a spurious residual floor no Newton iteration can remove.

Appendix B. Degree-of-freedom accounting, worked

Equation (9), $M + K = n_{A,\text{free}} + n_\alpha$, is an assertion rather than a derived quantity (Section 2.3); this appendix works it for the reference configuration and states the general rule it specialises.

The reference case. At $n = 8$ per side there are 9 coefficients per surface and 18 in total. The prescribed-leading-edge treatment removes 2 of these from the free set: $A_{u,0}$ and $A_{l,0}$ are geometrically fixed to the reference fit's own values rather than being Newton unknowns, leaving $n_{A,\text{free}} = 16$. Because they are prescribed directly rather than tied by an equality row, no separate shared-leading-edge constraint row is needed for this configuration, so $K = 0$. The angle of attack is fixed for the self-consistency protocol, removing the camber-incidence null direction of Section 5.1, so $n_\alpha = 0$. Equation (9) therefore requires $M = 16 + 0 - 0 = 16$ target-station equations, exactly the QR-pivoted station count reported throughout Sections 4 and 5.1, and exactly the value recorded in the run's own accounting record.

The general rule. Restating Eq. (9) in full generality: for n Bernstein coefficients per surface, hence $2(n+1)$ raw coefficients across two surfaces, with $c_{LE} \in \{0, 1\}$ constraint rows tying the leading-edge coefficients, used only when the leading edge is left free rather than prescribed; $p_{LE} \in \{0, 2\}$

coefficients removed by prescription; and any additional geometric constraint rows K_{extra} from trailing-edge thickness, inscribed area or curvature (Section 2.2),

$$n_{A,\text{free}} = 2(n + 1) - p_{LE}, \quad K = c_{LE} + K_{\text{extra}}, \quad M = n_{A,\text{free}} + n_\alpha - K, \quad (15)$$

which is Eq. (9) rearranged, not a different statement. This is the isolated-airfoil specialisation of the rigid-body bookkeeping quoted in Section 2.3: the three overall modes of scale, rotation and translation are not exposed as free Newton unknowns in this implementation, since chord, position and rotation are fixed by convention rather than solved for, so they contribute neither to $n_{A,\text{free}}$ nor to M .

The rule’s lineage is Lighthill’s own well-posedness bookkeeping [Lighthill, 1945]: a conformal-mapping inverse solve is well-posed only when the number of free shape parameters exactly matches the number of independent closure conditions, contour closure, leading-edge radius and circulation, that the target and geometry family jointly admit. Equation (9) is that same counting argument, discretised for a finite-dimensional CST basis and asserted programmatically rather than checked by hand.

Appendix C. The onset-closure Jacobian rows

Section 2.6 describes the forced-transition onset closure narratively; this appendix states its Jacobian-row structure explicitly, since it is the one residual block in this paper’s extended system that is not simply reused unchanged from the host solver.

At a forced-transition onset station with endpoint states U_1 upstream, still formally laminar-indexed, and U_2 downstream, freshly turbulent, the vendor three-row station residual is $R = [R_{\text{mom}}, R_{\text{shape}}, R_{\text{lag}}]$. Rows 0 and 1, momentum and shape parameter, are the vendor’s own ordinary turbulent closure evaluated directly across $[x_1, x_2]$, an unchanged code path, so their state and geometry Jacobian blocks are exactly the vendor’s own, reused verbatim. Row 2, the shear-lag equation, is replaced with the onset condition $U_2[2] - c_{ttr}(U_2) = 0$, where c_{ttr} is the same transition-correlation shear-stress function the vendor’s own transition update uses to initialise a newly turbulent node.

Because c_{ttr} is a function of the downstream state alone, specifically of θ , the kinematic shape parameter, the momentum-thickness Reynolds number and the edge velocity at that node, its Jacobian row has the structure

$$R_U[2, 0:4] = 0 \quad (\text{no } U_1 \text{ dependence}), \quad R_U[2, 4:8] = e_{sa} - \frac{\partial c_{ttr}}{\partial U_2} \bigg|_{U_2}, \quad R_x[2, :] = 0, \quad (16)$$

where $e_{sa} = [0, 0, 1, 0]$ selects the shear-lag component of U_2 and $\partial c_{ttr}/\partial U_2$ is the vendor’s own analytic derivative block, already computed by vendor code for the identical purpose inside its transition update: reused, not re-derived. The row’s independence from U_1 and from the geometry is not an approximation. It follows structurally from c_{ttr} ’s own functional form, verified by direct inspection of the vendor routine, and it is what makes this replacement row well-defined at a station where the natural closure’s internal transition-point solve has no solution.

Appendix D. Convergence-order estimation and the residual floor

Section 5.8 reports local convergence orders, and the estimates are sensitive enough to the estimation rule that the rule belongs in the record.

Given a residual sequence $r_0, r_1, \dots$, the three-point local order is

$$p_k = \frac{\log(r_k/r_{k-1})}{\log(r_{k-1}/r_{k-2})}, \quad (17)$$

which follows from assuming $r_k \approx C r_{k-1}^p$ and eliminating C between two consecutive steps. It is exact only asymptotically and is notoriously noisy on short sequences, which is why Section 5.8 reports the spread as well as the median.

Three rules govern its application here. First, the estimator is evaluated on the last three residuals lying strictly above 10^{-9} . That threshold is the finite-difference floor of Section 2.4: with a step of 10^{-6} and double precision, each Jacobian column carries a relative error of order 10^{-10} , so residuals at or below that magnitude reflect the Jacobian’s own noise rather than the iteration’s progress, and an order computed across them is meaningless. Runs reaching the floor in fewer than three post-initial steps are reported as unscorable rather than being given a number, which is why seven of the twelve converged runs carry no entry in Table 14. Second, the raw convergence-order field stored with each run does *not* apply this exclusion and is never quoted; every order in this paper is recomputed from the stored residual history under the rule above. Third, the residual block the estimator is applied to is stated explicitly, because it matters: for the ablation-matrix runs it is the full extended residual $\| [R; T - C p_{target}; GA - b] \|$, and for the self-consistency run of Section 4 it is the target block alone, because the flow block is already converged at that run’s starting point and contributes only its own floor.

Appendix E. Reproducibility and software environment

The repository at <https://github.com/SharathSPhD/CINS> carries the engine, every configuration file, the archived run records, the figure scripts, the test suite and the source of this manuscript. The flow solver `vendor/mfoil/mfoil.py` [Fidkowski, 2022] is vendored unmodified under the MIT licence and is never edited; all solver access passes through a single adapter module, which is what makes the forced-transition closure of Section 2.6 implementable without a fork.

Every run writes an immutable record under `experiments/results/` carrying the commit identifier, a hash of the fully-resolved configuration, the random seed, fixed at 42 throughout, and enough metadata to regenerate the run. No number in this paper is hand-entered; every table entry traces to a stored record, and Appendix F names the record for each table and figure.

Regeneration is by module. The ablation-matrix figures, being the convergence overlay, the order sweep, the cost bar chart and the station-selection comparison, come from `cins.benchmarks.figures`. The geometry, pressure, flow-solution and panel-gallery figures come from `cins.benchmarks.paper_figures`, and the formulation figures from `cins.benchmarks.paper_figures_theory`; both re-run the underlying monolithic solves fresh rather than reading cached scalars, because the stored records hold scalar summaries and not the coefficient vectors or field distributions themselves. The cost, convergence and panel-outcome figures come from `cins.benchmarks.paper_figures_archive`, which reads only stored records and re-runs nothing. Individual runs are reproduced through `experiments/run_t7.py` and `cins.benchmarks.runner`, and the test suite through `pytesttests/`. The one qualification on the data-availability statement is therefore that figures are regenerable from an archived *configuration* plus its recorded seed and commit, which for four of the figure families means re-running the solve rather than replotting stored arrays.

The environment is Python 3.11 or later with NumPy 1.26 or later, SciPy 1.11 or later, Matplotlib 3.8 or later and Pydantic 2.5 or later, as pinned in the project metadata; the vendored solver is the 2023-06-28 release. All timings quoted in Section 5.7 were measured on a single machine under a

single-threaded configuration and are reported for their ratio rather than their absolute value, which is hardware-dependent.

Appendix F. Figure and table provenance

Table 17 maps every figure and table in this manuscript to the record it was produced from, so that no source path need appear in a caption or in running prose. Paths are relative to the repository root. Entries marked as constructed are diagrams authored in the manuscript source rather than generated from data.

Table 17: Provenance of every figure and table.

Float	Source record
Fig. 1, 2	experiments/results/t8/figures/paper/, from paper_figures_theory
Fig. 3, 4, 17	constructed in the manuscript source
Fig. 5	experiments/results/t8/figures/paper/fig_viscous_solution.png
Fig. 6	experiments/results/t8/figures/h3_convergence_overlay.png
Fig. 7, 8, 9	experiments/results/t8/figures/paper/, from paper_figures
Fig. 10, 12	experiments/results/t8/figures/, from cins.benchmarks.figures
Fig. 11	experiments/results/t8/figures/paper/fig_le_identifiability.png
Fig. 13	experiments/results/t8/panel_*/result.json
Fig. 14, 15, 16	experiments/results/t8/uiuc_panel_summary.json; experiments/results/t8/*/result.json; configs/experiments/panel_uiuc/*.yaml
Tab. 1, 2, 3	formulation and instrumentation, no run record
Tab. 4	experiments/results/t7_naca2412/result.json; residual history from the companion diagnostics.json
Tab. 5	experiments/results/t8/t8_n08_baseline/, t8_station_even/
Tab. 6	experiments/results/le_stations/result.json
Tab. 7	experiments/results/t8/t8_n{04,06,08,10,12}*/; t2_gram_conditioning.json
Tab. 8, 9	experiments/results/t8/t8_*/result.json
Tab. 10, 11, 12	experiments/results/t8/uiuc_panel_summary.json
Tab. 13	experiments/results/t8/t8_n08_baseline*/, t8_init_perturbed/, control_cold_control/
Tab. 14	residual histories in experiments/results/t8/*/result.json, recomputed per Appendix D
Tab. 15, 16	rows (i)–(ii) as above; rows (iii)–(vi) from the cited literature

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